Open Mock Test No. 1

TOPICS OF THE TEST

Complete Syllabus of NEET

MM : 720

OPEN MOCK TEST - 1

Time : 3 Hrs. 20 Min.

[PHYSICS]

Choose the correct answer :

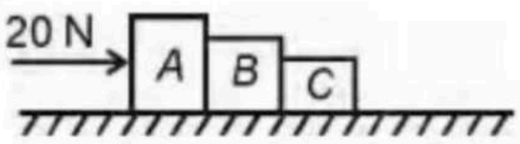
SECTION-A

1. Among the following which one is the weakest fundamental force?
   (1) Electromagnetic force (2) Gravitational force
   (3) Weak nuclear force (4) Strong nuclear force

2. The dimensions of a wooden block are 1.1 m × 2.36 m × 3.1 m. The volume of wooden block to the correct significant figure is
   (1) 8.0 (2) 8.04
   (3) 8.0476 (4) 8.047

3. Two projectiles *A* and *B* are thrown from the same point with velocities u and $\frac{u}{2}$ respectively. If *B* is thrown at an angle 45° with horizontal. What is the angle of projection of *A* with the horizontal if their horizontal range are same?
   (1) $\sin^{-1}\left(\frac{1}{4}\right)$ (2) $\frac{1}{2}\sin^{-1}\left(\frac{1}{4}\right)$
   (3) $2\sin^{-1}\left(\frac{1}{4}\right)$ (4) $\frac{1}{2}\sin^{-1}\left(\frac{1}{8}\right)$

4. Three blocks *A*, *B* and *C* of masses 5 kg, 3 kg and 2 kg respectively are in contact on a frictionless surface as shown in figure. If a force of 20 N is applied on block *A*, then ratio of contact force between *A* and *B* to the contact force between *B* and *C* will be

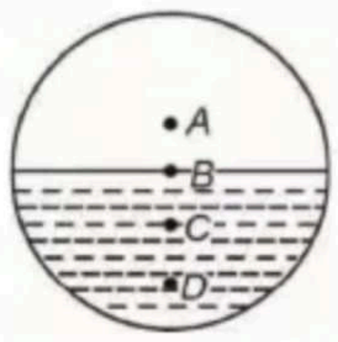


   (1) $\frac{3}{2}$ (2) $\frac{1}{2}$
   (3) $\frac{5}{2}$ (4) 1

5. A steel sphere of mass m_1 moving with velocity of 5 m/s collide with another particle of mas m_2 elastically moving in same direction with velocity 2 m/s. The velocity of the particle after the collision is ($m_1 >>> m_2$)
   (1) 10 m/s (2) 4 m/s
   (3) 5 m/s (4) 8 m/s

6. Three identical spheres each of radius '*R*' are placed touching each other so that their centres *A*, *B* and *C* lie on a straight line. The position of their centre of mass from centre of *A* is
   (1) $\frac{2R}{3}$ (2) $2R$
   (3) $\frac{5R}{3}$ (4) $\frac{4R}{3}$

7. A hollow sphere is half filled with water. Which of the following points, indicate the position of centre of mass of the system as shown in figure?



   (1) *A* (2) *B*
   (3) *C* (4) *D*

Space for Rough Work

8. A particle is executing SHM such that at mean position it has total energy of 24 J and has potential energy 4 J, then time-average of its K.E. is

(1) 5 J (2) 10 J

(3) 20 J (4) 15 J

9. When a satellite is shifted from a nearer orbit to a farther orbit, its

(a) Gravitational potential energy increases

(b) Kinetic energy increases

(c) Gravitational potential energy decreases

(d) Kinetic energy decreases

Choose the correct option.

(1) Only (a) is correct

(2) (a) and (d) are correct

(3) (a) and (c) are correct

(4) (a) and (b) are correct

10. The work done in stretching a wire by 0.2 mm is 8 J. What will be the work done in stretching another wire of same material, but with double the radius and half the length by 0.2 mm?

(1) 32 J (2) 16 J

(3) 64 J (4) 48 J

11. A large open container of negligible mass and uniform cross-sectional area has a small hole of area *a* in its side wall near the bottom. The container is kept over a smooth horizontal floor and contains a liquid of density ρ. Assuming that the liquid starts flowing through the hole, the acceleration of the container will be (*A* is area of cross-section of container)

(1) $\frac{2ag}{A}$ (2) $\frac{ag}{A}$

(3) $\frac{2Ag}{a}$ (4) $\frac{Ag}{a}$

12. The coefficient of cubical expansion of water is negative between

(1) 273 K and 277 K (2) 0 K and 4 K

(3) 300 K and 304 K (4) 4°C to 10°C

13. If γ denotes the ratio of two molar specific heats of a gas, the ratio of slopes of isothermal and adiabatic *P-V* curves at their point of intersection for the monoatomic gas will be $\left(\gamma = \frac{C_p}{C_v}\right)$

(1) $\frac{3}{2}$ (2) $\frac{2}{3}$

(3) $\frac{5}{3}$ (4) $\frac{7}{5}$

14. $Q = W$ is true for (symbol have their usual meaning)

(1) Isochoric process (2) Adiabatic process

(3) Isobaric process (4) Isothermal process

15. Match the following.

(A)	Doppler effect	(P)	Depth of oceans
(B)	Beats	(Q)	Velocity of stars
(C)	Echo	(R)	Standardisation of frequency

(1) A – Q, B – R, C – P (2) A – R, B – P, C – Q

(3) A – P, B – Q, C – R (4) A – P, B – R, C – Q

16. A transverse wave of amplitude 0.25 m, wavelength 2 m and frequency 4 Hz is propagating in a string along positive *x*-direction. The equation of this wave is

(1) $y = 0.25 \sin(\pi x - 8\pi t)$

(2) $y = 0.25 \sin(\pi x + 8\pi t)$

(3) $y = 0.25 \sin(4\pi x - 4\pi t)$

(4) $y = 0.25 \sin(4\pi x + 4\pi t)$

Space for Rough Work

17. The speed of sound in air at STP is 300 m/s. If the air pressure becomes double, the temperature remaining the same, the speed of sound would become

(1) 600 m/s (2) $300\sqrt{2}$ m/s

(3) 300 m/s (4) 1200 m/s

18. Two charges, each of 50 μC but opposite in sign, are placed 4 cm apart. The electric field intensity at a point which is at a distance of 4 cm from the mid-point of the dipole on the axial line.

(1) 10^8 N/C (2) 10^9 N/C

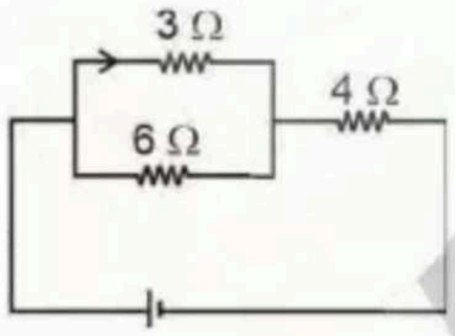
(3) 10^7 N/C (4) 10^{10} N/C

19. Six charges of same magnitude q are placed at the six corners of hexagon of side a. The value of electric potential at the centre of the hexagon will be $\left(k = \frac{1}{4\pi\varepsilon_0}\right)$

(1) $\frac{kq}{a}$ (2) $\frac{3\sqrt{3}\,ka}{a}$

(3) $\frac{3kq}{a}$ (4) $\frac{6kq}{a}$

20. If the current through 3 Ω resistor is 2 A, then power dissipated in 4 Ω resistor will be



(1) 18 W (2) 36 W

(3) 12 W (4) 48 W

21. In a potentiometer circuit a cell of e.m.f. 3 V gives balancing point at 45 cm length of wire. If another cell of e.m.f. 2 V replaces the first cell, then at what length of the wire, the balancing point occurs?

(1) 30 cm (2) 20 cm

(3) 60 cm (4) 40 cm

22. Magnetic induction at the centre of a current carrying circular loop of area π square metre is 0.2 Tesla. The magnetic moment of the loop is (μ_0 is the permeability of air)

(1) $\frac{0.1\pi}{\mu_0}$ (2) $\frac{0.2\pi}{\mu_0}$

(3) $\frac{0.4\pi}{\mu_0}$ (4) $\frac{0.3\pi}{\mu_0}$

23. A circular wire loop of radius r rotates about the y-axis with angular velocity ω. The normal to the loop is always perpendicular to the y-axis. At time $t = 0$, the normal is parallel to the z-axis. An external magnetic field $\vec{B} = B_y\hat{j} + B_z\hat{k}$ is applied. The e.m.f. (ε) induced in the loop is

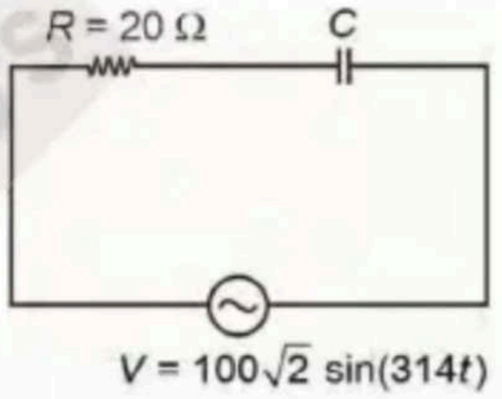
(1) $\pi r^2\omega B_y \sin\omega t$

(2) $\pi r^2\omega B_z \cos\omega t$

(3) $\pi r^2\omega B_z \sin\omega t$

(4) $\pi r^2\omega B_y \cos\omega t$

24. In the given R-C circuit, if the voltage across the resistor is 80 volt, then the capacitance is nearly



(1) 53 μF

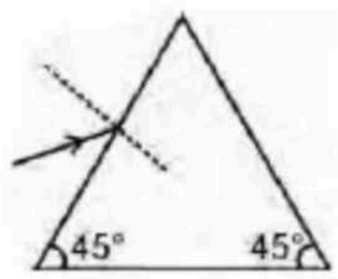
(2) 212 μF

(3) 112 μF

(4) 2 μF

Space for Rough Work

25. The minimum deviation produced by the prism having refractive index $\sqrt{\frac{3}{2}}$ as shown in the figure is



(1) 60° (2) 45°

(3) 15° (4) 30°

26. Distance of an object from the first focus of an equi-convex lens is 20 cm and distance of its real image from second focus is 80 cm. The focal length of the lens is

(1) 20 cm (2) 60 cm

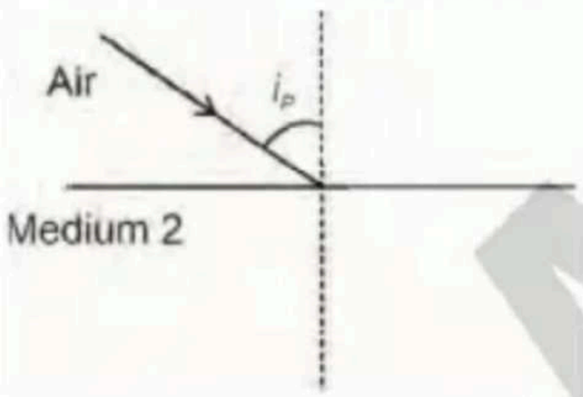
(3) 40 cm (4) 30 cm

27. In Young's double slit experiment, 50 fringes are observed in the central view zone with light of wavelength 4000 Å. The number of fringes that will be observed in the same view zone with the light of wavelength 5000 Å is

(1) 60 (2) 50

(3) 30 (4) 40

28. The tangent of polarizing angle according to Brewster's law is numerically equal to



(1) Mass density of the medium 2

(2) Refractive index of the medium 2

(3) Velocity of light in medium 2

(4) Elastic modulus of medium 2

29. An electromagnetic radiation has an energy 14.4 keV. To which region of electromagnetic spectrum does it belong?

(1) Infrared region

(2) Visible region

(3) Microwave region

(4) *x*-ray region

30. What should be the approximate velocity of an electron so that its momentum becomes equal to that of a photon of wavelength 4800 Å?

(1) 1000 m/s (2) 1500 m/s

(3) 700 m/s (4) 100 m/s

31. The nuclei count observed from a radioactive source at t second was N_0 and at $2t$ second it was $\frac{N_0}{4}$. The nuclei count observed at $\frac{7t}{2}$ second will be

(1) $\frac{N_0}{32}$ (2) $\frac{N_0}{16}$

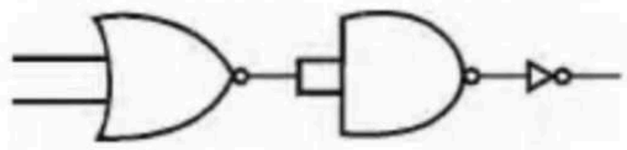
(3) $\frac{N_0}{8}$ (4) $\frac{N_0}{64}$

32. The time period of electron in the ground state of hydrogen atom is three times the time period of the electron in the second excited state of a certain hydrogen like atom (atomic number Z). The value of Z is

(1) 4 (2) 2

(3) 3 (4) 9

33. The circuit given below is equivalent to



(1) NOR gate (2) OR gate

(3) AND gate (4) NAND gate

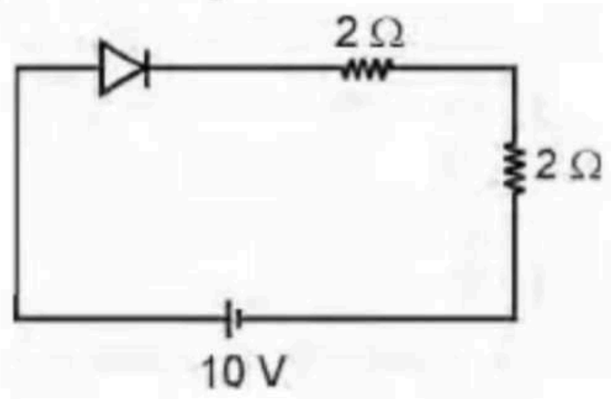
Space for Rough Work

34. In a common emitter configuration of a transistor, the voltage drops across a 400 Ω resistor in the collector circuit is 0.4 V when the collector supply voltage is 4 V. If the current gain in common base mode is 0.98, the base current is

(1) $\frac{1}{30}$mA (2) $\frac{1}{60}$mA

(3) $\frac{1}{49}$mA (4) $\frac{1}{98}$mA

35. In the given circuit for ideal diode, the current through the battery is



(1) 2 A (2) 2.5 A

(3) 1 A (4) 1.5 A

SECTION-B

36. What are the respective number of α and β^- particles emitted in the following radioactive decay?

$$^{200}_{90}X \longrightarrow {}^{172}_{82}Y$$

(1) 7 and 6 (2) 8 and 6

(3) 6 and 8 (4) 8 and 8

37. The radius of curvature of a convex mirror is 80 cm. When a real object is placed at *A* its image is formed at *B*. If the size of image is half that of the object, then distance between *A* and *B* is

(1) 20 cm

(2) 30 cm

(3) 40 cm

(4) 60 cm

38. Two electric bulbs whose resistances are in the ratio of 1 : 3 are connected in parallel to a constant voltage source. The power dissipated in them has the ratio

(1) 2 : 1 (2) 3 : 1

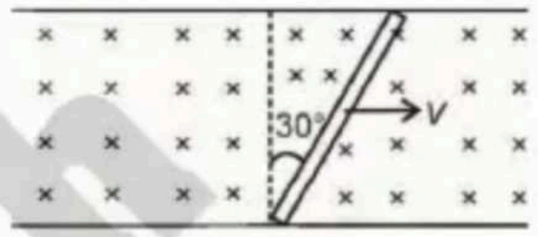
(3) 1 : 4 (4) 1 : 9

39. A magnetising field of 5000 A/m produces a magnetic flux of 5×10^{-5} weber in an iron rod. If the area of cross-section of the rod is 0.5 cm^2, then permeability of the rod will be (in SI units)

(1) 1×10^{-3} (2) 2×10^{-4}

(3) 3×10^{-5} (4) 4×10^{-6}

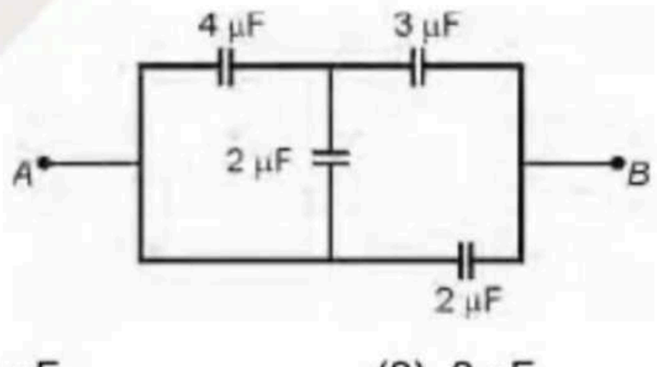
40. A rod of length ℓ slides on two long horizontal rails as shown in the figure. The emf induced between the ends of the rod is



(1) $\frac{\sqrt{3}B v \ell}{2}$ (2) $\frac{B v \ell}{2}$

(3) $\sqrt{3}\,B v \ell$ (4) $B v \ell$

41. Four capacitors are connected in a circuit as shown in figure. The effective capacitance between point *A* and *B* will be



(1) 4 μF (2) 2 μF

(3) 6 μF (4) 1 μF

Space for Rough Work

42. Two equal negative charges $-q$ are fixed at the points $(0, a)$ and $(0, -a)$ on the y-axis. A positive charge Q is released from rest from the point $(2a, 0)$ on the x-axis. The charge will

(1) Execute S.H.M about the origin

(2) Moves to the origin and remains at rest

(3) Moves to infinity

(4) Execute oscillatory motion but not S.H.M

43. A car turns a corner on a horizontal road at a constant speed of 10 m/s. If the coefficient of friction is 0.5, the minimum radius of the turn (in metre) so that car can turn safely is

(1) 20 m

(2) 10 m

(3) 18 m

(4) 36 m

44. A stone of mass 10 kg is lying at the bed of a large lake 10 m deep. If the relative density of the stone is 2, the amount of work done by external force to bring slowly it at the surface of the lake will be (neglect all resistive forces)

(1) 490 J (2) 245 J

(3) 385 J (4) 770 J

45. A homogeneous disc with a radius 0.1 m and mass 4 kg rotates about an axis passing through its centre and perpendicular to the plane in the influence of a tangential force. The angular velocity of the disc as a function of time is given by the formula $\omega = (4 + 3t)$ rad/s. The tangential force applied to the rim of the disc is

(1) 1.2 N

(2) 1.8 N

(3) 0.9 N

(4) 0.6 N

46. Surface tension of a liquid is numerically equal to

(1) Force on the surface

(2) Surface energy

(3) Potential energy

(4) Surface energy per unit area

47. Two identical rods of same material, transfer a given amount of heat in 16 second, when they are joined in series. If they are joined in parallel, then they will transfer same heat in same condition in time

(1) 4 s (2) 32 s

(3) 48 s (4) 64 s

48. A particle performs simple harmonic motion about $x = 0$ with an amplitude a and time period T. Speed of the particle at $x = \frac{a}{2}$ will be

(1) $\frac{\pi a}{T}$ (2) $\frac{3\pi^2 a}{T}$

(3) $\frac{\pi a\sqrt{3}}{T}$ (4) $\frac{\pi a\sqrt{3}}{2T}$

49. The equation of stationary wave along a stretched string is given by $y = 5\sin\frac{\pi x}{3}\cos 40\pi t$ (where x and y are in cm and t is in second).

The separation between consecutive node and antinode will be

(1) 1.5 m (2) 3 m

(3) 2 m (4) 4 m

50. A ball is dropped freely from large height while another is thrown vertically downward with an initial velocity 'v' from the same point, simultaneously. After 't' second they are separated by a distance of

(1) $\frac{vt}{2}$ (2) $\frac{gt^2}{2}$

(3) vt (4) $vt + \frac{1}{2}gt^2$

Space for Rough Work

[CHEMISTRY]

SECTION-A

51. A metal crystallises with a face-centred cubic lattice. If edge length of unit cell is 420 pm then the radius of the metal atom will be

(1) 148.5 pm (2) 182 pm

(3) 210 pm (4) 121.3 pm

52. Mass of concentrated nitric acid solution required to prepare 400 mL of 3 M HNO_3 solution is (The concentrated nitric acid is 70% $\left(\frac{w}{w}\right)$ HNO_3)

(1) 60 g (2) 82 g

(3) 142 g (4) 108 g

53. 14.2 g of Na_2SO_4 is dissolved in 200 g of water. If dissociation of salt in the solution is 90% then the freezing point of the solution will be

(K_f of water = 1.86 K kg mol^{-1})

(1) –1.8°C (2) –3.6°C

(3) –2.6°C (4) –4.1°C

54. The solubility of AgBr having solubility product 5×10^{-13} in 0.1 M $CaBr_2$ solution will be

(1) 5×10^{-12} M (2) 2.5×10^{-12} M

(3) 2.2×10^{-6} M (4) 5×10^{-10} M

55. Which among the following contains highest number of atoms?

(1) 8 g of CH_4 (2) 16 g of SO_2

(3) 9 g of $C_6H_{12}O_6$ (4) 22 g of CO_2

56. **Statement I:** Ionic radius of O^{2-} is greater than F^-.

Statement II: O^{2-} and F^- are isoelectronic species.

In light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

57. For He^+ ion, the de Broglie wavelength of electron in third Bohr orbit is

[Given that Bohr radius, a_0 = 52.9 pm]

(1) 79.4π pm (2) 158.7π pm

(3) 211.6π pm (4) 92.4π pm

58. The angular momentum of an electron in 3*p* orbital is equal to

(1) $2\sqrt{3}h$ (2) $\sqrt{3}h$

(3) $\sqrt{6}h$ (4) $\sqrt{2}h$

59. Pair of isostructural molecules among the following is

(1) BCl_3 and ClF_3 (2) XeO_3 and SO_3

(3) NH_3 and PCl_3 (4) SF_4 and XeF_4

60. Density of O_2 gas at 127°C and 4.1 atm pressure will be (R = 0.082 L atm K^{-1} mol^{-1})

(1) 3.2 gL^{-1} (2) 4 gL^{-1}

(3) 1.5 gL^{-1} (4) 2.5 gL^{-1}

61. If rate constant of a reaction is $9.212 \times 10^{-3}s^{-1}$ then the time required for the completion of 20% of the reaction will be

(1) 200 s (2) 100 s

(3) 50 s (4) 25 s

Space for Rough Work

62. Match the name of the ores given in column I with composition given in column II

	Column I		Column II
(a)	Malachite	(i)	Fe_2O_3
(b)	Calamine	(ii)	$FeCO_3$
(c)	Siderite	(iii)	$CuCO_3.Cu(OH)_2$
(d)	Haematite	(iv)	$ZnCO_3$

The correct match is

(1) (a) – (iii), (b) – (iv), (c) – (i), (d) – (ii)

(2) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(4) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

63. Which among the following complex ions has highest spin only magnetic moment value?

(1) $[Co(H_2O)_6]^{3+}$

(2) $[MnCl_6]^{3-}$

(3) $[Fe(CN)_6]^{3-}$

(4) $[Mn(CN)_6]^{3-}$

64. Correct order of boiling point of the given compounds is

(1) HI > HF > HBr > HCl

(2) HF > HI > HBr > HCl

(3) HF > HCl > HBr > HI

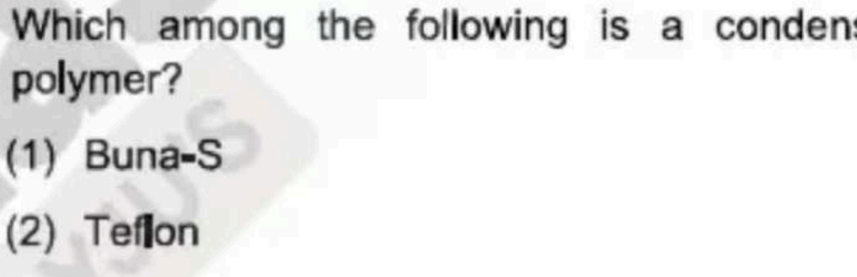
(4) HI > HF > HCl > HBr

65. An aqueous solution of $Ni(NO_3)_2$ is electrolysed between platinum electrodes using a current of 3.86 amperes for 100 minutes. The mass of nickel deposited at the cathode is (Atomic mass of nickel = 58.7 u)

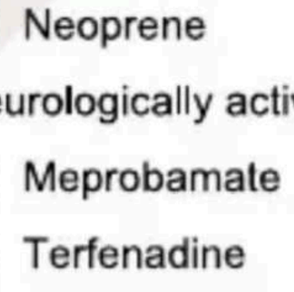
(1) 16.2 g (2) 3.5 g

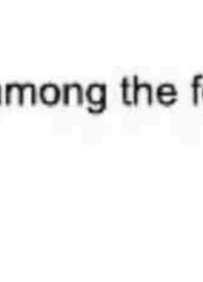
(3) 7 g (4) 10 g

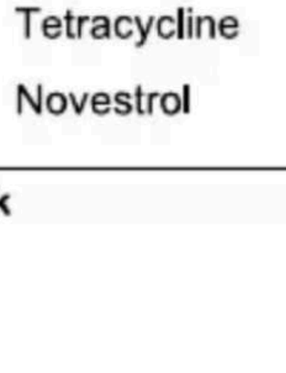
66. Consider the following reaction sequence



Major product C is

(1) 

(2) 

(3) 

(4)

67. Which among the following is a condensation polymer?

(1) Buna-S

(2) Teflon

(3) Terylene

(4) Neoprene

68. Neurologically active drug among the following is

(1) Meprobamate

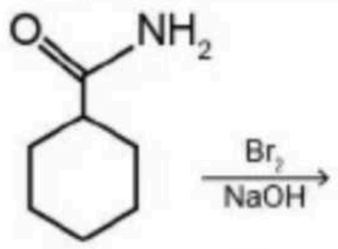
(2) Terfenadine

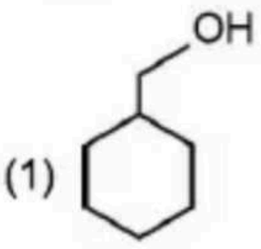
(3) Tetracycline

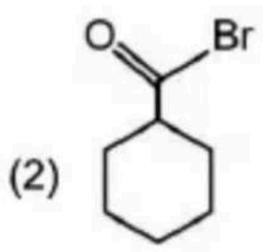
(4) Novestrol

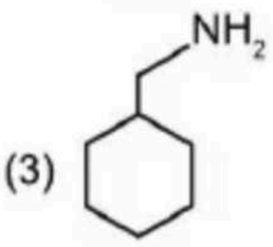
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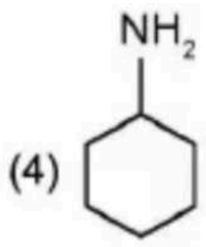
69. Major product of the given reaction is



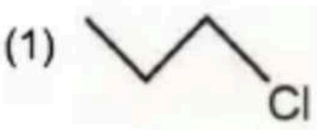
(1) 

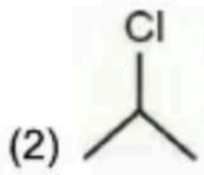
(2) 

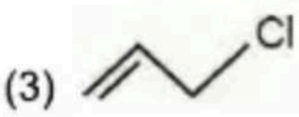
(3) 

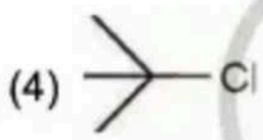
(4) 

70. Which among the following will react fastest by S_N1 mechanism?

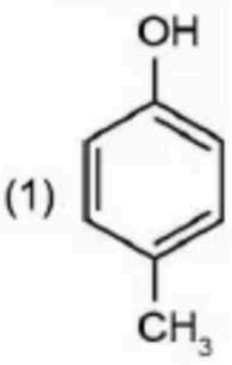
(1) 

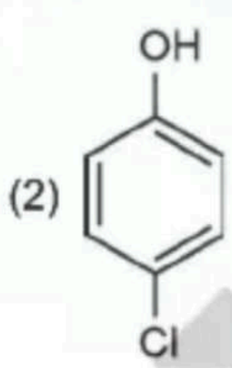
(2) 

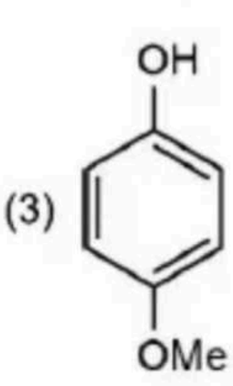
(3) 

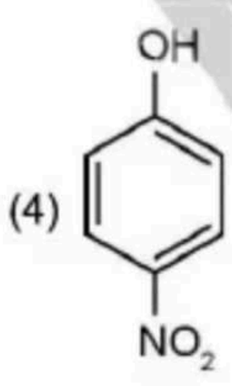
(4) 

71. Most acidic compound among the following is

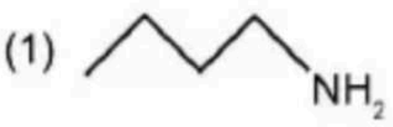
(1) 

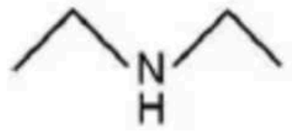
(2) 

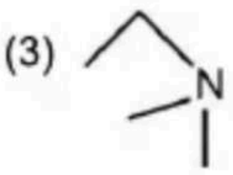
(3) 

(4) 

72. Compound A having molecular formula $C_4H_{11}N$ on reaction with benzenesulphonyl chloride gives B which dissolves in aqueous alkali. Compound A could be

(1) 

(2) 

(3) 

(4) 

73. Statement I: Sucrose on hydrolysis gives equimolar mixture of D-(+)-glucose and D-(-) fructose.

Statement II: Sucrose reduces Tollens' reagent.

In light of the above statements, choose the correct answer.

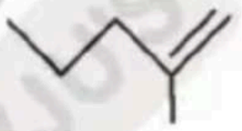
(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

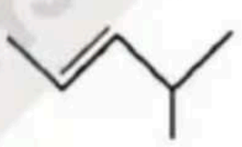
(3) Statement I is correct but statement II is incorrect

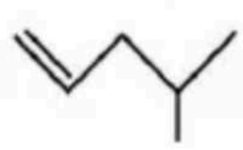
(4) Statement I is incorrect but statement II is correct

74. Which among the following will liberate highest amount of energy on catalytic hydrogenation?

(1) 

(2) 

(3) 

(4) 

75. In sulphur estimation, 0.5 g of an organic compound gave 0.699 g of barium sulphate. The percentage of sulphur in the compound is (Molar mass of $BaSO_4$ = 233 g mol^{-1})

(1) 27.5% (2) 19.2%

(3) 42.5% (4) 10.5%

Space for Rough Work

76. The pair of compounds which can be distinguished by NaOI is

(1) Ethanol and Ethanal

(2) Acetone and Benzaldehyde

(3) Acetophenone and Ethanol

(4) Benzaldehyde and Propanal

77. Which among the following is not formed in the given reaction?

$CH_3CHO + CH_3CH_2CHO \xrightarrow[(ii)\ \Delta]{(i)\ NaOH}$

(1) $CH_3 - CH = CH - CHO$

(2) $CH_3 - CH_2 - \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} = CH - CHO$

(3) $CH_3 - CH = \underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}} - CHO$

(4) $CH_3 - CH_2 - CH = CHCHO$

78. Which among the following ions is colourless in aqueous medium?

(1) Cu^{2+} (2) V^{3+}

(3) Sc^{3+} (4) Ni^{2+}

79. Which among the following is tetrahedral in shape and diamagnetic in nature?

(1) $[PtCl_4]^{2-}$ (2) $[Ni(CO)_4]$

(3) $[Cu(NH_3)_4]^{2+}$ (4) $[Pt(NH_3)_2Cl_2]$

80. The standard Gibbs energy change for the given cell reaction will be

(Given: $E^{\circ}_{Al^{3+}/Al} = -1.66\ V$ and $E^{\circ}_{Cu^{2+}/Cu} = 0.34\ V$)

$2Al(s) + 3Cu^{2+}(aq) \rightarrow 2Al^{3+}(aq) + 3Cu(s)$

(1) $-579\ kJ\ mol^{-1}$ (2) $-1158\ kJ\ mol^{-1}$

(3) $-386\ kJ\ mol^{-1}$ (4) $-965\ kJ\ mol^{-1}$

81. The gases obtained on the reaction of copper and zinc with dilute nitric acid respectively are

(1) NO_2 and N_2O (2) N_2O and N_2

(3) N_2O and NO (4) NO and N_2O

82. A gas is allowed to expand in a well insulated container against a constant external pressure of 5 atm from an initial volume of 1.25 L to a final volume of 3.25 L. The change in internal energy ΔU of the gas will be

(1) –625 J (2) –882 J

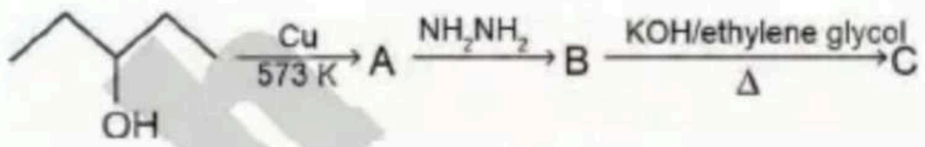
(3) – 1311 J (4) –1013 J

83. Maximum permissible limit of lead in drinking water is about

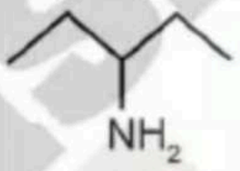
(1) 150 ppb (2) 100 ppb

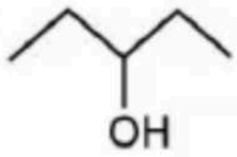
(3) 200 ppb (4) 50 ppb

84. Consider the following reaction sequence.



Major product C is

(1) 

(2) 

(3) 

(4) 

85. Match the compounds given in column I with their chemical nature given in column II.

	Column I		Column II
(a)	CO	(i)	Amphoteric
(b)	Na_2O	(ii)	Neutral
(c)	Cl_2O_7	(iii)	Basic
(d)	Al_2O_3	(iv)	Acidic

Space for Rough Work

Choose the correct match.

(1) (a) – (ii), (b) – (iii), (c) – (iv), (d) – (i)

(2) (a) – (iii), (b) – (ii), (c) – (iv), (d) – (i)

(3) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

(4) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

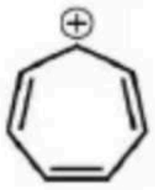
SECTION-B

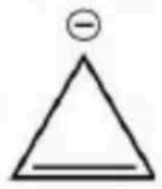
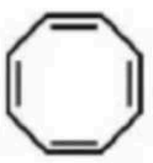
86. Ratio of root mean square speed of O_2 and He at 27°C will be

(1) 1 : 4 (2) $1:\sqrt{2}$

(3) $1:2\sqrt{2}$ (4) 1 : 2

87. Which among the following is an aromatic species?

(1)  (2) 

(3)  (4) 

88. Monomer of orlon is

(1) Chloroprene

(2) Styrene

(3) Acrylonitrile

(4) Vinyl chloride

89. Consider the following reaction at equilibrium

$$N_2(g)+3H_2(g)\rightleftharpoons 2NH_3(g)$$

If small amount of helium gas is added to the above equilibrium at constant pressure and temperature then

(1) Formation of ammonia increases

(2) Dissociation of ammonia increases

(3) There is no effect on equilibrium

(4) Equilibrium constant value decreases

90. Which among the following species, does not show disproportionation reaction?

(1) ClO^- (2) ClO_4^-

(3) ClO_3^- (4) ClO_2^-

91. If volume strength of H_2O_2 is 22.4 V then $\left(\frac{w}{V}\right)\%$ of H_2O_2 in the solution will be

(1) 3.4% (2) 6.8%

(3) 9.8% (4) 13.6%

92. Match the elements given in column I with their flame colour given in column II.

	Column I		Column II
(a)	Sodium	(i)	Apple green
(b)	Caesium	(ii)	Brick red
(c)	Calcium	(iii)	Blue
(d)	Barium	(iv)	Yellow

The correct match is

(1) (a) – (ii), (b) – (iii), (c) – (i), (d) – (iv)

(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

(3) (a) – (iii), (b) – (iv), (c) – (ii), (d) – (i)

(4) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

93. **Assertion (A):** Aqueous solution of borax is acidic in nature.

Reason (R): Borax on reaction with aqueous hydrochloric acid gives boric acid as one of the products.

In light of the above statements, choose the correct answer.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(2) Both (A) and (R) are correct and (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

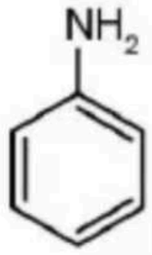
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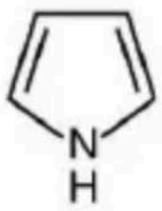
94. Which among the following is most easily soluble in water?

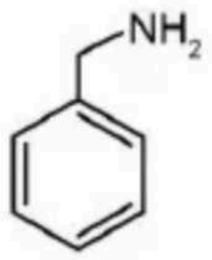
(1) $BaSO_4$ (2) $SrSO_4$

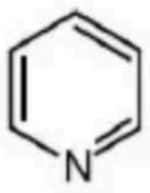
(3) $CaSO_4$ (4) $MgSO_4$

95. Most basic compound among the following is

(1) 

(2) 

(3) 

(4) 

96. If the rate constant of the reaction, $X \rightarrow Y$ is 2.5×10^{-4} mol L^{-1} s^{-1} and initial concentration of X is 3 M then the concentration of Y after 100 minutes will be

(1) 0.5 M (2) 1.0 M

(3) 1.5 M (4) 2.0 M

97. pH of 0.2 M solution of sodium acetate will be (pKa of acetic acid = 4.76)

(1) 5.0 (2) 6.5

(3) 8.5 (4) 9.0

98. 400 mL of 0.5 M HCl solution undergoes neutralisation reaction with 600 mL of 0.2 M NaOH solution. The heat released during neutralisation process is

(1) 6.85 kJ (2) 57.1 kJ

(3) 11.42 kJ (4) 20.5 kJ

99. Copper sol is coagulated most easily by

(1) Al^{3+} (2) Ba^{2+}

(3) SO_4^{2-} (4) PO_4^{3-}

100. Which among the following is not formed on the reaction of white phosphorus with thionyl chloride?

(1) PCl_3 (2) $POCl_3$

(3) SO_2 (4) S_2Cl_2

[BOTANY]

SECTION-A

101. Read the following statements w.r.t Mendel's cross-hybridisation experiments using pure pea lines

(a) A pure line undergoes continuous self-pollination, shows stable trait inheritance and expression for several generations.

(b) Mendel considered seven characters for his experiments.

(c) Mendel selected 7 true-breeding plants for his experiments.

Choose the **correct** statement(s).

(1) Only (a) and (b) (2) Only (b) and (c)

(3) Only (c) and (a) (4) Only (c)

102. Pairing of homologous chromosomes occurs in which stage of cell division?

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

103. Choose the **correct** option w.r.t location of root hairs.

(1) Root cap

(2) Zone of maturation

(3) Zone of elongation

(4) Zone of meristematic activity

Space for Rough Work

104. Heterostyly is an outbreeding device adopted by some plants. It is a type of
(1) Pollination method called cleistogamy
(2) Morphological method to prevent self-pollination
(3) Physiological self incompatibility
(4) Self pollination method

105. Which of the following statements is true for one turn of Krebs cycle?
(1) One molecule of $FADH_2$ is synthesized
(2) Three molecules of $NADH + H^+$ are oxidised to NAD^+
(3) Three molecules of $FADH_2$ are oxidised to FAD^+
(4) Only one molecule of NAD^+ is reduced to $NADH + H^+$

106. DNA-dependent DNA polymerase
(1) Catalyses polymerisation only in 5′ → 3′ direction
(2) Has polymerization rate of 50 nucleotides/second
(3) Has polymerization rate of 20,000 bp per second
(4) Transcribes the DNA for mRNA synthesis

107. Match the following columns w.r.t. type of interactions and their examples.

	Column I		Column II
a.	Mycorrhiza	(i)	Predation
b.	Eagle-Rat	(ii)	Amensalism
c.	*Penicillium* – Bacteria	(iii)	Parasitism
d.	Lice – human	(iv)	Mutualism

Select the **correct** option from the following:
(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(i), c(ii), d(iii)
(3) a(iv), b(iii), c(ii), d(i) (4) a(ii), b(iv), c(i), d(iii)

108. Xylem translocates
(1) Only water
(2) Inorganic nitrogen, water and hormones only
(3) Organic nitrogen, water, hormones and mineral salts
(4) Organic nitrogen, water, mineral salts and photosynthates

109. Read the following statements.
(a) The system of classification based on evolutionary relationships among various organisms was proposed by Linnaeus.
(b) *Eudorina* shows anisogamy.
(c) Laminarin is the chief storage food in Phaeophyceae members.
(d) Haplontic life cycle is prevalent in mosses.
How many of the above statement(s) is/are **correct**?
(1) 4 (2) 3
(3) 2 (4) 1

110. Choose the **correctly** matched pair:
(1) Photorespiration – Wasteful process as it utilises ATP and $NADPH_2$
(2) Ruben and Kamen – Proved that O_2 liberated during photosynthesis comes from CO_2
(3) Tadpole like structure – Chlorophyll
(4) Kranz anatomy - Potato

111. When the centromere is present very close to one end of the chromosome, the chromosome is referred to as
(1) Metacentric
(2) Sub-metacentric
(3) Acrocentric
(4) Telocentric

Space for Rough Work

112. The transverse section of a plant part shows following anatomical features:
(a) Central pith.
(b) Conjoint and open vascular bundles.
(c) Medullary rays present in between the vascular bundles.
(d) Hypodermis is collenchymatous.
Identify the category of plant and its part
(1) Monocotyledonous stem
(2) Dicotyledonous stem
(3) Monocotyledonous root
(4) Dicotyledonous root

113. The function of NAD^+ in cellular respiration is
(1) To act as an electron carrier
(2) To act as an enzyme
(3) To act as an electron acceptor in ETS
(4) To produce ATP by providing electrons

114. Vegetative propagation in ginger is through
(1) Rhizome (2) Eye of tuber
(3) Leaf buds (4) Offset

115. Which one of the following is the "most time consuming and tedious process" w.r.t. plant breeding?
(1) Selection and testing of superior recombinants
(2) Testing, release and commercialisation of new cultivars
(3) Germplasm collection
(4) Cross-hybridisation among the selected parents

116. Choose the **correct** order of various stages in hydrarch succession.
(1) Forest → Zooplanktons → Phytoplanktons → rooted floating plants → rooted-submerged plants
(2) Phytoplanktons → submerged plants → submerged free-floating plants → reed-swamp → marsh-meadow → scrubs → forest
(3) Marsh-meadow → reed-swamp → trees → phytoplanktons → rooted floating plants → rooted – submerged plants
(4) Floating plants → rooted-submerged plants → trees → forest → Zooplankton → phytoplankton

117. ZW-ZZ type of sex determination is seen in
(1) Grasshopper (2) Honey Bee
(3) Birds (4) *Drosophila*

118. Match the following columns and select the **correct** option:

	Column I		**Column II**
a.	RER	(i)	Glycosylation
b.	SER	(ii)	Protein synthesis
c.	Golgi complex	(iii)	Spindle formation
d.	Centriole	(iv)	Lipid synthesis

(1) a(ii), b(i), c(iv), d(iii)
(2) a(ii), b(iv), c(i), d(iii)
(3) a(iii), b(ii), c(i), d(iv)
(4) a(i), b(ii), c(iv), d(iii)

119. Mortality refers to
(1) Birth rate
(2) Death rate
(3) Number of individuals leaving a habitat
(4) Number of individuals entering a habitat

120. The layer of an anther which is **not** protective in function is
(1) Tapetum (2) Middle layers
(3) Epidermis (4) Endothecium

121. In which of the following forms, copper is absorbed by plants?
(1) Cuprous ions (2) Cupric ions
(3) Free element (4) Cu^+

Space for Rough Work

122. The 'good' ozone that acts as a shield by absorbing UV radiation from the sun is present in which of the following layers of the atmosphere?

(1) Troposphere (2) Stratosphere

(3) Mesosphere (4) Ionosphere

123. The phytohormone which was first isolated from human urine is

(1) Gibberellic acid (2) Cytokinin

(3) Auxin (4) Ethylene

124. On Earth, which of the following regions have the greatest biodiversity?

(1) Western Ghats of India

(2) Polar regions

(3) Tropical Amazonian rain forests

(4) Indo-Burma Himalayas

125. Which of the following statements is/are correct about prions?

(a) Cause mad cow disease in cattle

(b) Size is similar to viruses

(c) Made up of RNA only

(d) Causes Cr-Jacob disease in humans

(1) All (a), (b), (c) and (d)

(2) (a), (b) and (d) only

(3) (c) only

(4) (b), (c) and (d) only

126. Which of the following is a filamentous algae?

(1) *Ulothrix* (2) *Chlamydomonas*

(3) *Volvox* (4) *Porphyra*

127. In dicot leaves, guard cells are

(1) Dumb-bell shaped (2) Kidney shaped

(3) Rectangular (4) Barrel shaped

128. Acetic acid is commercially obtained from

(1) *Penicillium*

(2) *Aspergillus niger*

(3) *Lactobacillus*

(4) *Acetobacter aceti*

129. In DNA fingerprinting, differences in some specific regions in DNA sequence are identified using a specific type of probes, which are

(1) VNTRs

(2) Repetitive RNA

(3) Non-polymorphic DNA

(4) Single nucleotides

130. In most ecosystems, the pyramids of number, energy and biomass are upright, **except**

(1) Pyramid of number in grassland

(2) Pyramid of biomass in grassland

(3) Pyramid of number in sea

(4) Pyramid of biomass in sea

131. Which of the following statement(s) is/are **correct** for cork cambium?

(a) It is also called phellogen.

(b) It is also called extra-stelar cambium.

(c) Differentiation of cork cambium cells form cork on the inner side of itself.

(d) The function of cork is nutritive.

(1) (a), (b) and (c) only

(2) All (a), (b), (c) and (d)

(3) (a) and (b) only

(4) (b), (c) and (d) only

Space for Rough Work

132. In the S-phase of interphase, **A** and **B** are duplicated.

A and B, respectively, are

(1) Centrioles in nucleus and DNA in cytoplasm

(2) DNA and proteins both in cytoplasm

(3) DNA in nucleus and centrioles in cytoplasm

(4) Proteins in nucleus and centrioles in cytoplasm

133. In *Antirrhinum*, a red flower was crossed with a white flower and in F_1 generation, pink flowers were obtained. When pink flowers were selfed, the F_2 generation showed red, white and pink flowers.

Choose the **correct** statement from the following:

(1) This experiment follows the principle of dominance

(2) Phenotypic ratio of F_2 is 1 : 2 : 1 (Red : Pink : White)

(3) Pink colour in F_1 is due to co-dominance

(4) Law of segregation is not appliable in this experiment

134. Read the following statements and choose the suitable option for A and B accordingly.

(a) A and B were not included in 5 kingdom classification given by Whittaker.

(b) B are both living and non-living in nature.

(c) A are the indicators of pollution.

(d) A show a mutualistic relationship.

The correct option for A and B respectively is

(1) Viruses and Bacteria

(2) Viruses and Lichens

(3) Lichens and Viruses

(4) Lichens and Bacteria

135. Choose the **incorrectly** matched pair from the following w.r.t. crop and its resistance to corresponding pest/disease.

(1) Pusa Gaurav – Aphids

(2) Himgiri – Hill bunt

(3) Pusa Swarnim – Bacterial blight

(4) Pusa Sadabahar – Chilly mosaic virus

SECTION-B

136. The contrasting characters present in a pair, used in a taxonomical aid, is referred to as

(1) Key (2) Lead

(3) Couplet (4) Manual

137. Match the components of column I with those of column II w.r.t. photosynthesis and choose the **correct** option.

	Column I		**Column II**
a.	Hatch and Slack pathway	(i)	Calvin cycle
b.	RuBP	(ii)	C_4 plants
c.	Cyclic photophosphorylation	(iii)	Photosystem I

(1) a(i), b(ii), c(iii) (2) a(ii), b(i), c(iii)

(3) a(iii), b(i), c(ii) (4) a(ii), b(iii), c(i)

138. Zoospores in *Chlamydomonas* and megaspore cell of the ovule are produced by

(1) Meiosis and mitosis respectively

(2) Mitosis

(3) Mitosis and meiosis respectively

(4) Meiosis

Space for Rough Work

139. Which one of the following statements is **correct** for plants pollinated by water?

(1) Most of the flowering plants show pollination by water

(2) All aquatic plants use water for pollination

(3) Water pollinated flowers are colourful and produce nectar

(4) The pollen grains of water-pollinated plants are protected by a mucilaginous covering

140. The World Summit on Sustainable Development held in 2002 at Johannesburg, South Africa, aimed at

(1) Reducing emission of greenhouse gases

(2) Protecting the stratosphere

(3) Taking appropriate measures for conservation of biodiversity and sustainable utilization of its benefits

(4) Significantly reducing the current rate of biodiversity loss at global, regional and local levels

141. Match List-I with List-II

	List-I		List-II
a.	Histone protein synthesis	(i)	Interphase
b.	Longest phase of cell cycle	(ii)	S-phase
c.	Longest stage of meiosis I	(iii)	Diakinesis
d.	Terminalisation of chiasmata	(iv)	Prophase I

Choose the **correct** option:

(1) a(iv), b(iii), c(i), d(ii) (2) a(ii), b(i), c(iv), d(iii)

(3) a(i), b(ii), c(iii), d(iv) (4) a(ii), b(iv), c(i), d(iii)

142. In a cross between a colourblind female and normal male, what percent of the progeny will be colourblind female?

(1) 50%

(2) 100%

(3) Zero

(4) 25%

143. Pick the **odd** one from the following statements.

(1) The terminal electron acceptor of ETS is oxygen

(2) The electrons are transported in the ETS by the electron carrier proteins

(3) The protons move from mitochondrial matrix to the cytoplasm during the movement of electrons in the ETS

(4) Complex V of the ETS is involved in ATP synthesis

144. "CO_2 is a major limiting factor in photosynthesis". Select the **incorrect** statement from the options given below.

(1) More than 0.05% of CO_2 can be damaging over long periods

(2) At low light conditions and high CO_2 concentration, C_3 plants show increased rate of photosynthesis

(3) At low light intensity and high CO_2 concentration, C_4 plants don't show increased rate of photosynthesis

(4) At high light intensity, C_3 plants show increased rate of photosynthesis

145. Leaf tendrils are present in

(1) Cactus

(2) *Pisum sativum*

(3) Members of Cucurbitaceae

(4) *Bryophyllum*

Space for Rough Work

146. In which genetic condition, each cell in the affected person has 44 + XO condition?
(1) Down's syndrome
(2) Klinefelter's syndrome
(3) Turner's syndrome
(4) Edwards syndrome

147. Statins, used as blood-cholesterol lowering agents, are produced by
(1) Fungus *Trichoderma polysporum*
(2) Bacterium *Streptococcus*
(3) Fungus *Monascus purpureus*
(4) Bacterium *Clostridium*

148. What is the role of RNA polymerase II in the process of transcription in eukaryotes?
(1) Transcribes 28S, 18S and 5.8S rRNAs
(2) Transcribes hnRNA
(3) Transcribes tRNA
(4) Transcribes snRNA

149. Identify the **correct** statement with regard to G_2 phase of the interphase.
(1) It is the interval between mitosis and initiation of DNA replication
(2) Tubulin protein synthesis occurs in it
(3) Chromosome duplicates in this phase
(4) Most of the organelles duplicate in this phase

150. Match the items in column I with those of column II and choose the **correct** option.

	Column I		Column II
a.	Gibberellins	(i)	Present in root and shoot apex
b.	Auxins	(ii)	Overcome apical dominance
c.	Abscissic acid	(iii)	Seed germination
d.	Cytokinin	(iv)	Induces dormancy

(1) a(iii), b(iv), c(i), d(ii) (2) a(ii), b(iii), c(i), d(iv)
(3) a(iii), b(i), c(iv), d(ii) (4) a(ii), b(i), c(iii), d(iv)

[ZOOLOGY]

SECTION-A

151. Select the cell junction that facilitates communication between adjacent cells for rapid transfer of ions and molecules.
(1) Tight junction
(2) Gap junction
(3) Adhering junction
(4) Macula adherens

152. Select the **incorrect** statement.
(1) hCG, hPL and relaxin are produced in women only during pregnancy
(2) During pregnancy, relaxin is produced by ovary
(3) Placenta also acts as an endocrine tissue and produces several hormones including estrogen and progesterone
(4) Trophoblast contains stem cells which have the potency to give rise to all the tissues and organs

153. The condition diuresis can be defined as
(1) Increase in protein excretion
(2) Increase in urine volume
(3) Decrease in urine volume
(4) Increased H^+ and K^+ secretion

Space for Rough Work

154. Select a fatty acid from the following.
(1) Acetic acid (2) Palmitic acid
(3) Glutamic acid (4) Aspartic acid

155. Bursting of RBCs to release toxin called haemozoin which causes high fever with chills are symptoms of
(1) Amoebiasis (2) Malaria
(3) Filariasis (4) Ringworms

156. A cow is administered hormones (FSH-like) to induce follicular maturation and superovulation in
(1) Artificial insemination
(2) MOET
(3) Interspecific hybridization
(4) Cross-breeding

157. The first clinical gene therapy was given in 1990 to a 4-year old girl with A deficiency. In some children, this deficiency can be cured by B .

Select the **correct** option to fill in the blanks respectively.
(1) ADA and *cry* gene
(2) ADA and bone marrow transplantation
(3) Insulin and enzyme replacement therapy
(4) Insulin and α–interferons

158. A chronic disorder characterised by damage to alveolar walls and decrease in respiratory surface is
(1) Asthma (2) Emphysema
(3) Diarrhoea (4) Muscular dystrophy

159. Choose the **odd** one w.r.t. axial skeleton.
(1) Atlas and axis
(2) Tibia and fibula
(3) Ethmoid and sphenoid
(4) Sternum and ribs

160. Select the **correct** set containing only diploid cells.
(1) Primary spermatocyte and ootid
(2) Spermatozoa and spermatogonia
(3) Spermatogonia and primary oocytes
(4) Spermatids and oogonia

161. Foetal ejection reflex triggers the release of X from maternal pituitary.

Choose the option which fills the blank 'X' correctly.
(1) Vasopressin (2) ACTH
(3) Oxytocin (4) ADH

162. Excessive use of antibiotics resulted in natural selection of antibiotic resistant varieties of bacteria in a much lesser time scale. This is an example of
(1) Adaptive radiation
(2) Founder's effect
(3) Anthropogenic action
(4) Saltation

163. Low pO_2, high pCO_2, high H^+ and high temperature are the conditions favourable for
(1) Association of O_2 to haemoglobin to form oxyhaemoglobin
(2) Dissociation of O_2 from oxyhaemoglobin within the lung's alveoli
(3) Dissociation of O_2 from oxyhaemoglobin in body tissues
(4) Shifting of oxygen dissociation curve to the left

164. A hormone releasing IUD among the following is
(1) Lippes loop
(2) LNG-20
(3) Diaphragms
(4) Multiload-375

Space for Rough Work

165. Choose the **incorrect** statement with respect to coagulation of blood.
(1) Platelets release certain factors which are involved in the clotting of blood
(2) Certain factors, released by the tissue at the site of injury can also initiate coagulation
(3) The excess in number of platelets above 3.5 lakh/mm^3 leads to clotting disorder which will lead to excessive loss of blood from the body
(4) Fibrins are formed by the conversion of inactive fibrinogen in the plasma by the enzyme thrombin

166. Select the **incorrect** statement.
(1) Cnidoblasts are used for anchorage, defense and for the capture of prey
(2) Ciliated comb plates help in locomotion in ctenophores
(3) Metameric segmentation is the characteristic feature of phylum Aschelminthes
(4) Choanocytes line the spongocoel and the canals in sponges

167. In a DNA of 500 nucleotide base pairs in length, cytosine is 40%. What will be the number of adenines?
(1) 400 (2) 100
(3) 500 (4) 200

168. Select the **correct** match.

(1)	Duct of gall bladder	Hepatic duct
(2)	Glisson's capsule	Mucosa of alimentary canal
(3)	Brunner's gland	Between common hepatic duct and duodenum
(4)	Sphincter of Oddi	Hepato-pancreatic duct

169. **Assertion (A):** Diaphragm, cervical caps and vaults are barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus.
Reason (R): Sterilization procedure is called vasectomy in the male and tubectomy in the female.
In the light of above statements, select the **correct** option.
(1) Both (A) and (R) are correct but (R) is not correct explanation of (A)
(2) Both (A) and (R) are correct and (R) is the correct explanation of (A)
(3) (A) is correct and (R) is incorrect
(4) (A) is incorrect and (R) is correct

170. Select the **odd** one w.r.t. radial symmetry in adult organisms of given phyla.
(1) Echinodermata (2) Cnidaria
(3) Ctenophora (4) Aschelminthes

171. Which of the following is **incorrect** w.r.t. Aves?
(1) Excretory organ is proboscis gland
(2) Presence of 4-chambered heart
(3) Presence of feathers and beak
(4) Presence of long bones which are hollow with air cavities

172. If a dsDNA is digested by a particular restriction endonuclease, a little away from the centre of palindromic sequence, what will happen?
(1) DNA fragments with flush ends will be produced
(2) Resultant DNA fragments may not have the overhanging stretches
(3) The DNA fragments will facilitate the action of DNA ligase
(4) The DNA fragments will not have sticky ends

Space for Rough Work

173. In 1983, Eli Lilly, an American company prepared two DNA sequences corresponding to A and B chains of human insulin and introduced them in plasmids of

(1) *Salmonella typhi*

(2) *Thermus aquaticus*

(3) *Meloidogyne incognita*

(4) *Escherichia coli*

174. Which of the following is **not** a method of diagnosis of a disease?

(1) PCR

(2) ELISA

(3) Radiography

(4) Radiotherapy

175. Read the following and choose the **incorrect** statement.

(1) rDNA technology serves the purpose of early diagnosis of diseases

(2) RTIs could lead to several complications like PID, still birth, ectopic pregnancy, cancer, *etc.*

(3) Hepatitis – B and HIV can also be transmitted by sharing of infected needles, surgical instruments, infected blood transfusion, *etc.*

(4) Mature insulin has three polypeptide chains

176. Which among the following belongs to class Mammalia?

(1) *Ascidia*

(2) *Canis*

(3) *Chelone*

(4) *Corvus*

177. Select the **odd** one w.r.t. secondary metabolites.

(1) Concanavalin A (2) Tyrosine

(3) Vinblastin (4) Ricin

178. During fertilization, a sperm comes in contact with the 'X' layer of the ovum and induces changes in the membrane that 'Y' the entry of additional sperms.

Select the **correct** option for 'X' and 'Y' respectively.

(1) Zona pellucida and block

(2) Zona pellucida and promote

(3) Corona radiata and promote

(4) Corona radiata and block

179. A person with which of the following blood group can receive the blood from universal donor only?

(1) A (2) O

(3) AB (4) B

180. Select an autoimmune disorder that affects neuromuscular junctions.

(1) Osteoporosis (2) Myasthenia gravis

(3) Gout (4) Muscular dystrophy

181. _____ are located in the wall of hollow visceral organs of the body.

Select the **correct** option to fill in the blank.

(1) Skeletal muscles (2) Smooth muscles

(3) Striated muscles (4) Cardiac muscles

182. Select the **odd** one w.r.t. functions of PTH.

(1) Increases levels of Ca^{2+} in blood

(2) Stimulates mineralisation of bones

(3) Increases Ca^{2+} absorption from digested food

(4) Stimulates Ca^{2+} reabsorption by renal tubules

183. All of the following are triploblastic coelomates **except**

(1) *Gorgonia* (2) *Limulus*

(3) *Pinctada* (4) *Nereis*

Space for Rough Work

184. Select the **incorrect** match.

(1) pO_2 in deoxygenated blood – 40 mmHg

(2) pO_2 in alveolar air – 104 mmHg

(3) pCO_2 in alveolar air – 45 mmHg

(4) pCO_2 in systemic arteries – 40 mmHg

185. Which of the following juices contains a variety of enzymes like maltase, dipeptidases and nucleosidases?

(1) Pancreatic juice

(2) Succus entericus

(3) Gastric juice

(4) Chyme

SECTION-B

186. In *Periplaneta americana*, a pair of large ovaries are present in

(1) 10th abdominal segment

(2) 2nd – 6th abdominal segments

(3) 6th – 7th abdominal segments

(4) 5th – 7th abdominal segments

187. Which one of the following statements is **correct** with respect to regulation of kidney functions?

(1) ANF is a vasoconstrictor and thereby decrease the blood pressure

(2) Renin converts the angiotensin II into angiotensin I

(3) Erythropoietin is produced by vasa recta of juxtamedullary nephrons

(4) When a normal person drinks a lot of water, ADH release is supressed

188. Read the following statements carefully and choose the option that is incorrectly explained.

(1) Proteins are chains of amino acids which are linked by peptide bonds

(2) Inulin is a polymer of fructose

(3) Starch can hold I_2 molecules in the helical portion

(4) The sugar found in polynucleotides is always ribose

189. Select the **correct** statement.

(1) The *tet*^{R} gene in pBR322 can be cleaved by *Pvu*I and *Pst*I

(2) One can ligate a foreign DNA at *Bam*HI site of tetracycline resistance gene in pBR322

(3) When a rDNA is inserted within the coding sequence of β-galactosidase, it results into insertional activation

(4) The rop codes for the polysaccharides involved in the replication of the plasmid

190. 'X' is called the functional unit of contraction and 'Y' is the anatomical unit of muscle.

Here, 'X' and 'Y' respectively are

(1) Sarcomere and muscle fibre

(2) Sarcomere and fascicles

(3) Fascicles and muscle fibre

(4) Fascia and muscle fibre

191. Consider the equation given below.

$S - G + S' \rightarrow S + S' - G$

The enzyme catalysing given reactions belongs to class

(1) I (2) III

(3) II (4) IV

Space for Rough Work

192. If the change in allele frequency occurs by chance, it is called

(1) Genetic equilibrium

(2) Genetic modification

(3) Genetic drift

(4) Gene flow

193. Who makes decisions regarding the validity of GM research and the safety of introducing GM – organisms for public services?

(1) European Foundation of Biotechnology

(2) European Federation of Biotechnology

(3) Genetic Engineering Approval Committee

(4) Genetic Engineering Appraisal Company

194. Read the statements carefully and select the **incorrect** one w.r.t. PCR.

(1) If the cycle is repeated 30 times, a DNA sample can be amplified a billion times

(2) The *Taq* polymerase adds nucleotides in $3' \rightarrow 5'$ direction to synthesize complementary strands

(3) Denaturation is required to separate dsDNA sample

(4) DNA polymerase is used in primer extension step

195. Read the following statements and select the **correct** option.

Statement A: The DNA fragments can be separated by a technique known as gel electrophoresis.

Statement B: The separated DNA fragments can be visualized only after staining the DNA with ethidium bromide followed by exposure of UV-radiation.

(1) Both statements A and B are correct

(2) Only statement A is incorrect

(3) Only statement B is incorrect

(4) Both statements A and B are incorrect

196. Find out the **correct** match between column I and column II.

	Column I		**Column II**
a.	Normal blood pressure	i.	0.5 seconds
b.	Duration of cardiac cycle	ii.	$\frac{120}{80}$ mmHg
c.	Duration between dub and lub	iii.	0.8 seconds

(1) a – ii, b – iii, c – i

(2) a – i, b – ii, c – iii

(3) a – iii, b – ii, c – i

(4) a – i, b – iii, c – ii

197. Read the statements carefully w.r.t. prostomium of *Pheretima* and choose **incorrect** one from the options given below.

(1) It is sensory in function

(2) A lobe which serves as covering for the mouth

(3) It is 1st body segment (buccal segment) which contains mouth

(4) It helps to open cracks in the soil into which it can crawl

Space for Rough Work

198. The floor of scala media is formed of

(1) Scala vestibuli

(2) Basilar membrane

(3) Reissner's membrane

(4) Tectorial membrane

199. In a population of humans, 90 out of 1000 babies are affected by a recessive condition called cystic fibrosis. In this population, the frequency of heterozygous individuals is

(1) 0.4

(2) 0.2

(3) 0.42

(4) 0.8

200. Select the **incorrect** pair.

(1)	Corpus callosum	–	Connection between cerebral hemispheres with a tract of nerve fibres
(2)	Association area	–	Contains a number of centres which control body temperature, urge for drinking
(3)	Hypothalamus	–	Contains several groups of neurosecretory cells
(4)	Medulla oblongata	–	Contains centres which control respiration, cardiovascular reflexes, *etc.*

❑ ❑ ❑

Aakash BYJU'S

Space for Rough Work

All India Aakash Test Series for NEET - 2023

OPEN MOCK TEST - 1 (Code-A)

Test Date : 26/03/2023

ANSWERS

1. (2)	41. (1)	81. (4)	121. (2)	161. (3)
2. (1)	42. (4)	82. (4)	122. (2)	162. (3)
3. (2)	43. (1)	83. (4)	123. (3)	163. (3)
4. (3)	44. (1)	84. (3)	124. (3)	164. (2)
5. (4)	45. (4)	85. (1)	125. (2)	165. (3)
6. (2)	46. (4)	86. (3)	126. (1)	166. (3)
7. (3)	47. (1)	87. (2)	127. (2)	167. (2)
8. (2)	48. (3)	88. (3)	128. (4)	168. (4)
9. (2)	49. (1)	89. (2)	129. (1)	169. (1)
10. (3)	50. (3)	90. (2)	130. (4)	170. (4)
11. (1)	51. (1)	91. (2)	131. (3)	171. (1)
12. (1)	52. (4)	92. (4)	132. (3)	172. (3)
13. (3)	53. (3)	93. (4)	133. (2)	173. (4)
14. (4)	54. (2)	94. (4)	134. (3)	174. (4)
15. (1)	55. (1)	95. (3)	135. (3)	175. (4)
16. (1)	56. (1)	96. (3)	136. (3)	176. (2)
17. (3)	57. (2)	97. (4)	137. (2)	177. (2)
18. (2)	58. (4)	98. (1)	138. (3)	178. (1)
19. (4)	59. (3)	99. (1)	139. (4)	179. (2)
20. (2)	60. (2)	100. (2)	140. (4)	180. (2)
21. (1)	61. (4)	101. (1)	141. (2)	181. (2)
22. (3)	62. (3)	102. (2)	142. (3)	182. (2)
23. (3)	63. (2)	103. (2)	143. (3)	183. (1)
24. (2)	64. (2)	104. (2)	144. (2)	184. (3)
25. (4)	65. (3)	105. (1)	145. (2)	185. (2)
26. (3)	66. (4)	106. (1)	146. (3)	186. (2)
27. (4)	67. (3)	107. (2)	147. (3)	187. (4)
28. (2)	68. (1)	108. (3)	148. (2)	188. (4)
29. (4)	69. (4)	109. (3)	149. (2)	189. (2)
30. (2)	70. (4)	110. (3)	150. (3)	190. (1)
31. (1)	71. (4)	111. (3)	151. (2)	191. (3)
32. (4)	72. (1)	112. (2)	152. (4)	192. (3)
33. (1)	73. (3)	113. (1)	153. (2)	193. (3)
34. (3)	74. (4)	114. (1)	154. (2)	194. (2)
35. (2)	75. (2)	115. (4)	155. (2)	195. (1)
36. (1)	76. (2)	116. (2)	156. (2)	196. (1)
37. (4)	77. (2)	117. (3)	157. (2)	197. (3)
38. (2)	78. (3)	118. (2)	158. (2)	198. (2)
39. (2)	79. (2)	119. (2)	159. (2)	199. (3)
40. (1)	80. (2)	120. (1)	160. (3)	200. (2)

HINTS & SOLUTIONS

[PHYSICS]

SECTION-A

1. Answer (2)

Hint and Sol.: $F_{SN} : F_E : F_{WN} : F_G = 1 : 10^{-2} : 10^{-13} : 10^{-39}$

2. Answer (1)

Hint: In multiplication or division, the final result should retain only that many significant figures as are there in the original number with the least number of significant figures.

Sol.: Volume = 1.1 m × 2.36 m × 3.1 m

$\approx 8.0476\ m^3$

$= 8.0\ m^3$

3. Answer (2)

Hint: $R = \dfrac{u^2 \sin 2\theta}{g}$

Sol.: $R_1 = R_2$

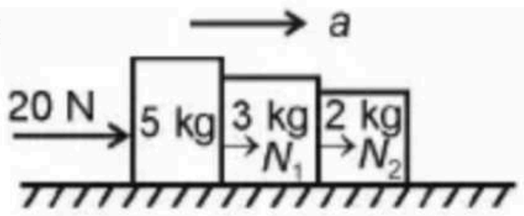
$$\frac{u_1^2}{g} = \frac{u_2^2 \sin 2\theta}{g}$$

$$\frac{u^2}{4} = u^2 \sin 2\theta \Rightarrow \theta = \frac{1}{2}\sin^{-1}\left(\frac{1}{4}\right)$$

4. Answer (3)

Hint: $F = ma$

Sol.:



$F = ma$

$$a = \frac{20}{5+3+2}$$

$= 2\ m/s^2$

$N_1 = (3 + 2)\ a$

$= 10\ N$

$N_2 = 2 \times a$

$= 4\ N$

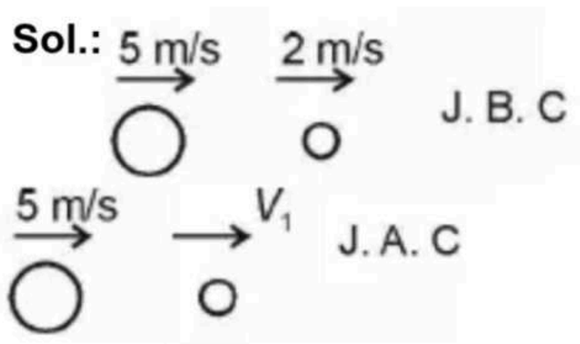
$$\frac{N_1}{N_2} = \frac{10}{4}$$

$$\frac{N_1}{N_2} = \frac{5}{2}$$

5. Answer (4)

Hint: Coefficient of restitution = $\dfrac{\text{Velocity of separation}}{\text{Velocity of approach}}$

Sol.:



$$e = \frac{v_1 - 5}{5 - 2}$$

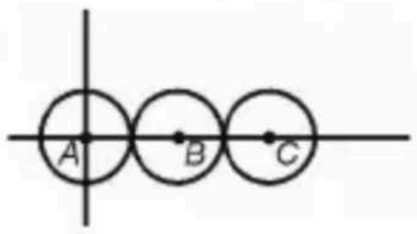
$$1 = \frac{v_1 - 5}{3}$$

$V_2 = 8\ m/s$

6. Answer (2)

Hint: $\vec{r}_{cm} = \dfrac{m_1\vec{r}_1 + m_2\vec{r}_2 + m_3\vec{r}_3}{m_1 + m_2 + m_3}$

Sol.:



$$X_{cm} = \frac{mx0 + m \times 2R + mx4R}{m + m + m}$$

$X_{cm} = 2R$

7. Answer (3)

Hint & Sol.: Mass of lower half of sphere is more therefore, centre of mass shift towards lower half.

8. Answer (2)

Hint: $(K.E.)_{max} = T.E. - P.E.$

Sol.: $(K.E.)_{max} = 24 - 4$

$= 20\ J$

$$\text{Average (K.E.)} = \frac{(K.E.)_{max}}{2}$$

$$= \frac{20}{2}$$

$= 10\ J$

9. Answer (2)

Hint & Sol.: When a satellite is lifted from lower orbit to a higher orbit then its gravitational potential energy increases and kinetic energy of satellite decreases.

10. Answer (3)

Hint: $W = \frac{1}{2} Y \left(\frac{\Delta \ell}{\ell}\right)^2 A\ell$

Sol.: $W_1 \propto \frac{A_1}{\ell_1}$

$W_2 \propto \frac{A_2}{\ell_2}$

$\frac{W_1}{W_2} = \frac{A_1}{\ell_1} \times \frac{\ell_2}{A_2}$

$\frac{W_1}{W_2} = \frac{A_1}{\ell_1} \times \frac{\ell_1}{2 \times 4A_1}$

$\frac{W_1}{W_2} = \frac{1}{8}$

$W_2 = 8 \times 8$

$= 64$ J

11. Answer (1)

Hint: Force on container $= \rho a v^2$

Sol.: $v = \sqrt{2gh}$

Mass of liquid in the container

$m_0 = \rho Ah$

$F = \rho a\, 2gh$

$m_0\, a_1 = \rho a\, 2gh$

$\rho Aha_1 = 2\rho\, agh$

$a_1 = \left(\frac{2ag}{A}\right)$

12. Answer (1)

Hint & Sol.: The coefficient of cubical expansion of water is negative between the temperature 273 K and 277 K.

13. Answer (3)

Hint: $\left|\left(\frac{dP}{dV}\right)_{\text{adiabatic}}\right| = -\gamma \frac{P}{V}$

Sol.: $\left(\frac{dP}{dV}\right)_{\text{isothermal}} = -\frac{P}{V}$

$\frac{\left(\frac{dP}{dV}\right)_{\text{adiabatic}}}{\left(\frac{dP}{dV}\right)_{\text{isothermal}}} = \gamma$

For monoatomic gas $\gamma = \left(\frac{5}{3}\right)$

14. Answer (4)

Hint & Sol.: In isothermal process temperature remains constant therefore $\Delta U = 0$

$Q = W + \Delta U$

$Q = W$

15. Answer (1)

Hint & Sol.: In motion of star, doppler effect causes change in the observed frequency of light. Beats is used in the standardization of frequency and to measure the depth of ocean concept of echo is used.

16. Answer (1)

Hint: Equation of propagating wave moving in positive x-direction of $y = A \sin(kx - \omega t)$.

Sol.: $A = 0.25$

$\omega = 2\pi f$

$= 8\pi$

$K = \frac{2\pi}{\lambda}$

$= \frac{2\pi}{2}$

$= \pi$

$y = 0.25A\sin(\pi x - 8\pi t)$

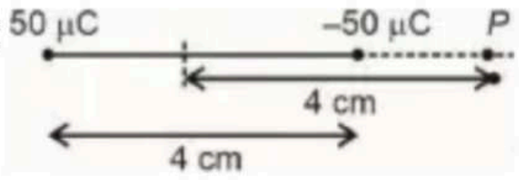
17. Answer (3)

Hint and Sol.: The speed of sound in air is independent of pressure of the gas. It only depends on temperature of gas.

18. Answer (2)

Hint: $E = \frac{kq}{r^2}$

Sol.:



$E_P = E_- + E_+$

$= \frac{k(50\,\mu C)}{(2\,cm)^2} - \frac{k(50\,\mu C)}{(6\,cm)^2}$

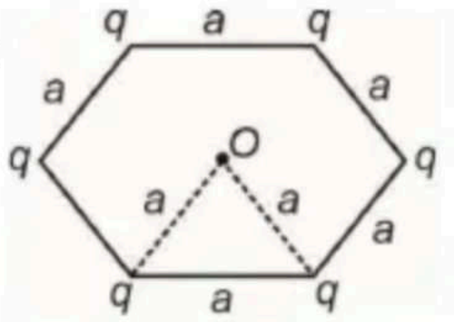
$= \frac{9 \times 10^9 \times 50 \times 10^{-6}}{4 \times 10^{-4}} - \frac{9 \times 10^9 \times 50 \times 10^{-6}}{36 \times 10^{-4}}$

$= 10^9$ N/C

19. Answer (4)

Hint: $V = V_1 + V_2 + V_3 \ldots\ldots V_6$

Sol.:



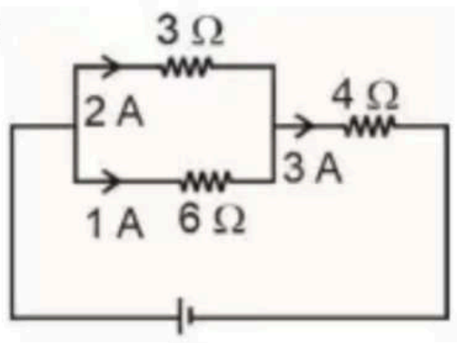
$$V = \left(\frac{kq}{a}\right) \times 6$$

$$= \frac{6kq}{a}$$

20. Answer (2)

Hint: $P = I^2R$

Sol.:



$P = I^2R$

$= (3)^2 \times 4$

$= 36$ W

21. Answer (1)

Hint: $E \propto \ell$

Sol.: $\frac{E_1}{E_2} = \frac{\ell_1}{\ell_2}$

$\frac{3}{2} = \frac{45}{\ell_2}$

$\ell_2 = 30$ cm

22. Answer (3)

Hint: $M = NIA$

Sol.: $B = \frac{\mu_0 i}{2R}$

$0.2 = \frac{\mu_0 i}{2R}$

$\Rightarrow \quad \frac{0.4}{\mu_0} = \frac{i}{R}$

$\Rightarrow \quad \pi R^2 = \pi$

$R = 1$

$M = \frac{0.4\pi}{\mu_0}$

23. Answer (3)

Hint: $\phi = \vec{B} \cdot \vec{A}$ and $\varepsilon = -\frac{d\phi}{dt}$

Sol.: $\phi = B_z \pi r^2 \cos\omega t$

$\varepsilon = -\frac{d\phi}{dt}$

$= \frac{-d}{dt}\left(B_z \pi r^2 \cos\omega t\right)$

$\varepsilon = B_z \omega \pi r^2 \sin\omega t$

24. Answer (2)

Hint: $X_c = \frac{1}{\omega C}$

Sol.: $V = 100 \sin(314t)$

$\omega = 314$

$V_R = IR$

$I = \frac{80}{20}$

$I = 4$ A

$\sqrt{V_R^2 + V_C^2} = 100$

$(80)^2 + V_C^2 = (100)^2$

$V_C^2 = (100)^2 - (80)^2$

$V_C = 60$

$V_C = I X_C$

$60 = 4 \times \frac{1}{\omega C}$

$C = \frac{4}{60 \times 314}$

$= 212\ \mu$F

25. Answer (4)

Hint: For minimum deviation $i = e$, $r_1 = r_2$

Sol.: By Snell's law

$\frac{\sin i}{\sin r} = \mu$

$r_1 = r_2 = r$

$r_1 + r_2 = A$

$r = \frac{A}{2}$

$= 45°$

$\frac{\sin i}{\sin 45°} = \frac{\sqrt{3}}{\sqrt{2}}$

$$\sin i = \frac{\sqrt{3}}{2}$$

$i = 60°$

$i + e = A + \delta$

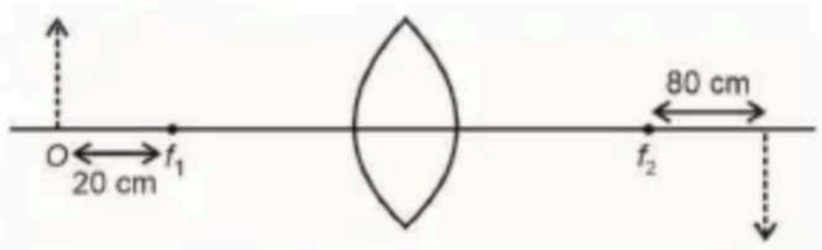
$60° + 60° = 90° + \delta$

$\delta = 30°$

26. Answer (3)

Hint: Use $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

Sol.:



$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{80+f} + \frac{1}{20+f} = \frac{1}{f}$

$(20 + f + 80 + f)f = (80 + f)(20 + f)$

$100f + 2f^2 = 1600 + 100f + f^2$

$f^2 = 1600$

$f = 40$ cm

27. Answer (4)

Hint: Fringe width $= \frac{\lambda D}{d}$

Sol.: $\frac{n_1\lambda_1 D}{d} = \frac{n_2\lambda_2 D}{d}$

$n_2 = \frac{n_1\lambda_1}{\lambda_2}$

$= \frac{50 \times 4000}{5000}$

$= 40$

28. Answer (2)

Hint & Sol.: $\mu = \tan(i_P)$

29. Answer (4)

Hint: $E = \frac{hc}{\lambda}$

Sol.: $\lambda = \frac{hc}{E}$

$= \frac{12400}{14.4 \times 10^3}$

$\approx$ 0.86 Å belongs to *x*-ray region.

30. Answer (2)

Hint: Momentum of photon $= \frac{h}{\lambda}$

Sol.: $mv = \frac{h}{\lambda}$

$v = \frac{h}{\lambda \times m}$

$= \frac{6.6 \times 10^{-34}}{4800 \times 10^{-10} \times 9.1 \times 10^{-31}}$

$\approx$ 1500 m/s

31. Answer (1)

Hint: $N = N_0 e^{-\lambda t}$

Sol.: $N_0 = N_1 e^{-\lambda t}$ ...(i)

$\frac{N_0}{4} = N_1 e^{-2\lambda t}$ ...(ii)

(i) ÷ (ii)

$4 = e^{\lambda t}$

$\lambda t = \ln 4$

$N = N_1 e^{-\lambda\left(\frac{7t}{2}\right)}$

$N = N_1 e^{-\frac{7}{2}\ln 4}$

$N = \frac{N_1}{e^{\ln 2^7}}$

$N = \frac{N_1}{2^7}$

$N = \frac{N_0 2^2}{2^7}$

$N = \frac{N_0}{2^5}$

$= \frac{N_0}{32}$

32. Answer (4)

Hint: $T \propto \frac{n^3}{Z^2}$

Sol.: $T_1 = 3T_2$

$\frac{1^3}{1^2} = \frac{3(3)^3}{Z^2}$

$Z = 9$

33. Answer (1)

Hint and Sol.: $Y_1 = \overline{A+B}$

$Y = \overline{\overline{Y_1 \cdot Y_1}}$

$= Y_1$

$= \overline{A+B} \Rightarrow$ NOR gate

34. Answer (3)

Hint: $I_C = \beta I_B$

Sol.: Given $V_{R_C} = i_c R_c$

0.4 V = i_C (400)

$\Rightarrow i_C$ = 1 mA

$\because \alpha = 0.98$

$$\therefore \beta = \frac{\alpha}{1-\alpha} = \frac{0.98}{1-0.98} = 49$$

$$i_B = \frac{i_C}{\beta} = \frac{1}{49}\text{mA}$$

35. Answer (2)

Hint: $I = \frac{V}{R}$

Sol.: $I = \frac{10}{2+2}$

= 2.5 A

SECTION-B

36. Answer (1)

Hint & Sol.: ${}^{200}_{90}X \longrightarrow {}^{172}_{82}Y + n_1\alpha + n_2\beta^-$

${}^{200}_{90}X \longrightarrow {}^{172}_{82}Y + n_1\,{}^{4}_{2}He + n_2\beta^-$

On balancing the mass number and atomic number, we get

$n_1 = 7$ and $n_2 = 6$

37. Answer (4)

Hint: Use $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Sol.: $f = \frac{R}{2}$

$= \frac{80}{2}$

= 40 cm

$m = -\frac{v}{u} = \frac{h_i}{h_o}$

$-\frac{v}{u} = -\frac{1}{2}$

$|u| = 2\,|v|$

$\frac{1}{|v|} + \frac{1}{-|u|} = \frac{1}{f}$

$1 - \frac{|v|}{|u|} = \frac{|v|}{f}$

$|v| = \frac{f}{2} = 20$ cm

$|u| = 2|v| = 40$ cm

$|u| + |v| = 40 + 20 = 60$ cm

38. Answer (2)

Hint: $P = \frac{V^2}{R}$

Sol.: $P \propto \frac{1}{R}$

$\frac{P_1}{P_2} = \frac{R_2}{R_1} = \frac{3}{1}$

39. Answer (2)

Hint: $\phi = \vec{B}\cdot\vec{A}$

Sol.: $\phi = \mu HA$

$\mu = \frac{\phi}{HA}$

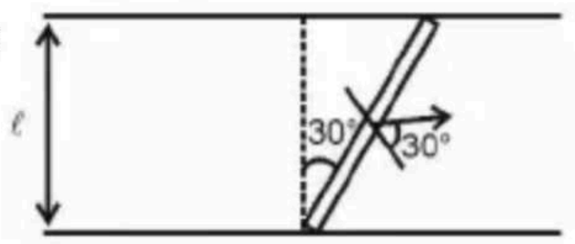
$= \frac{5\times10^{-5}}{5000\times\frac{1}{2}\times10^{-4}}$

= 2 × 10^{-4}

40. Answer (1)

Hint: $\varepsilon = \left(\vec{v}\times\vec{B}\right)\cdot\vec{\ell}$

Sol.:



$\varepsilon = Bv\ell\cos 30°$

$= \frac{Bv\ell\sqrt{3}}{2}$

41. Answer (1)

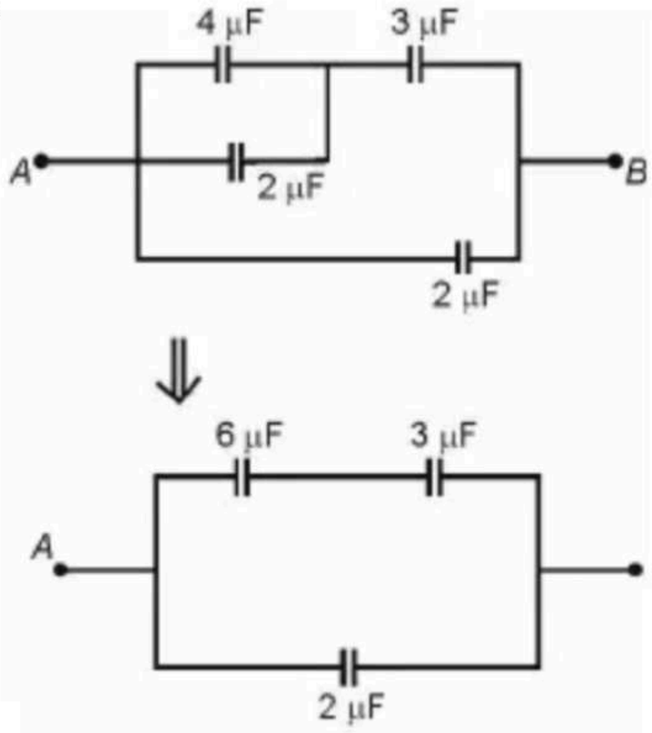
Hint: For parallel combination

$C_{eff} = C_1 + C_2 + \ldots C_n$

For series combination

$\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \ldots \frac{1}{C_n}$

Sol.:



$$C_{AB} = \frac{6\times3}{6+3} + 2$$

$= 4\ \mu F$

42. Answer (4)

Hint & Sol.: Charge will execute oscillatory motion but not S.H.M because electrostatic force acting on the charge is not proportional to displacement of charge particle from its mean position.

43. Answer (1)

Hint: Friction force = centripetal force

Sol.: $\mu mg = \frac{mv^2}{R}$

$$R = \frac{v^2}{\mu g}$$

$$= \frac{(10)^2}{0.5\times10}$$

$= 20\ m$

44. Answer (1)

Hint: Weight of the stone reduced due to upthrust force.

Sol.: ρ = density of the material of stone = $\frac{m}{V}$

$\therefore$ V = Volume of the stone

$$= \frac{m}{\rho} = \frac{10}{2\times10^3} = 5\times10^{-3}\,m^3$$

loss in weight = $V\sigma g$

$= 5 \times 10^{-3} \times 10^3 \times 9.8$

$= 49\ N$

Net weight to be lifted = $10 \times 9.8 - 49$

$= 49\ N$

$W = 49 \times 10$

$= 490\ J$

45. Answer (4)

Hint: $\tau = Fr_{\perp}$

Sol.: $\omega = 4 + 3t$

$$\alpha = \frac{d\omega}{dt}$$

$= 3\ rad/s^2$

$\tau = I \times \alpha$

$$F \times R = \frac{mR^2}{2}\alpha$$

$$F = \frac{mR}{2}\alpha$$

$$= \frac{4\times0.1\times3}{2}$$

$= 0.6\ N$

46. Answer (4)

Hint and Sol.: $S = \frac{dU}{dA}$

47. Answer (1)

Hint: $\frac{dQ}{dt} = \frac{T_1 - T_2}{R}$

Sol.: $\frac{Q}{t_1} = \frac{T_1 - T_2}{(2R)}$

R is thermal resistance of one rod

$$\frac{Q}{t_2} = \frac{T_1 - T_2}{\left(\frac{R}{2}\right)}$$

$$\frac{t_2}{t_1} = \frac{1}{4} \Rightarrow t_2 = 4\ s$$

48. Answer (3)

Hint: $v = \omega\sqrt{a^2 - x^2}$

Sol.: $v = \omega\sqrt{a^2 - x^2}$

$$x = \frac{a}{2}$$

$$v = \frac{\sqrt{3}a\,\omega}{2}$$

$$\omega = \frac{2\pi}{T}$$

$$v = \frac{\sqrt{3}a\pi}{T}$$

49. Answer (1)

Hint: Distance between consecutive node and antinode is $\frac{\lambda}{4}$

Sol.: $y = 5\sin\left(\frac{\pi x}{3}\right)\cos 40\pi t$

$\frac{2\pi}{\lambda} = \frac{\pi}{3}$

$\lambda = 6$

$\frac{\lambda}{4} = \frac{6}{4}$

= 1.5 m

50. Answer (3)

Hint: Relative acceleration of both the bodies will be zero.

Sol.: $\vec{S} = \vec{v}_{rel}\, t + \frac{1}{2}\vec{a}_{rel} \times t^2$

$\vec{a}_{rel} = 0$

$\vec{S} = v \times t$

[CHEMISTRY]

SECTION-A

51. Answer (1)

Hint: For fcc unit cell; $4r = \sqrt{2}a$

Sol.: $4r = \sqrt{2}a$

$r = \frac{\sqrt{2}}{4}a$

$r = \frac{\sqrt{2}}{4} \times 420$

r = 148.5 pm

52. Answer (4)

Hint: Mass of HNO_3 required = 400 × 3 × 10^{-3} × 63 = 75.6 g

Sol.: 70 g of HNO_3 is present in 100 g of concentrated nitric acid solution

75.6 g of HNO_3 is present in $\frac{100}{70} \times 75.6 = 108$ g

53. Answer (3)

Hint: $\Delta T_f = iK_f m$

Sol.:

	Na_2SO_4	$\rightleftharpoons$	$2Na^+$	+	SO_4^{2-}
Initially	1		0		0
Finally	(1 − 0.9)		0.9 × 2		0.9
	= 0.1		= 1.8		= 0.9

i(van't Hoff factor) = 0.1 + 1.8 + 0.9

i = 2.8

$\Delta T_f = iK_f m$

$\Delta T_f = 2.8 \times 1.86 \times \frac{14.2 \times 1000}{142 \times 200}$

$\Delta T_f = 2.6\,K$

Freezing point of solution = − 2.6° C

54. Answer (2)

Hint: Bromide of $CaBr_2$ (strong electrolyte) will act as common ion and will decreases the solubility of AgBr.

Sol.: $CaBr_2$ is strong electrolyte

$CaBr_2 \longrightarrow Ca^{2+} + 2Br^-$

$0.1 \qquad 2 \times 0.1 = 0.2$

$AgBr \rightleftharpoons Ag^+ + Br^-$

$(s) \qquad (0.2 + s) \simeq 0.2$

$[Ag^+][Br^-] = K_{sp}$

$s \times 0.2 = 5 \times 10^{-13}$

$s = \frac{5 \times 10^{-13}}{0.2}$

s = 2.5 × 10^{-12}

Solubility of AgBr = 2.5 × 10^{-12} M

55. Answer (1)

Hint: Number of atoms in molecule

$= \frac{\text{Mass of molecule}}{\text{Molar mass}} \times \text{atomicity} \times N_A$

Sol.:

- Number of atoms of $CH_4 = \frac{8}{16} \times 5 \times N_A$

  = 0.75 N_A
- Number of atoms of $SO_2 = \frac{16}{64} \times 3N_A$
- Number of atom of $C_6H_{12}O_6 = \frac{9}{180} \times 24N_A$

  $= 1.2N_A$

Number of atoms of $CO_2 = \frac{22}{44} \times 3N_A = 1.5N_A$

56. Answer (1)

Hint: Higher is the effective nuclear charge smaller is the size of ion.

Sol.:

- O^{2-} and F^- are isoelectronic species having 10 electrons each.
- Effective nuclear charge on F^- is more hence F^- is smaller in size than O^{2-}.

57. Answer (2)

Hint: $r_n = a_0 \frac{n^2}{Z}$

de Broglie wavelength λ is given as

$n\lambda = 2\pi r_n$

Sol.: $r_3 = a_0 \times \frac{(3)^2}{2}$

$r_3 = \frac{9}{2}a_0$

$\lambda = \frac{2\pi r_3}{3} \quad [\because n = 3]$

$\lambda = \frac{2\pi}{3} \times \frac{9}{2}a_0$

$\lambda = 3a_0\pi$

$\lambda = 3 \times 52.9\pi$ pm

$\lambda = 158.7\pi$ pm

58. Answer (4)

Hint: Angular momentum = $\sqrt{l(l+1)}h$

Sol.: For *p*-orbital l = 1

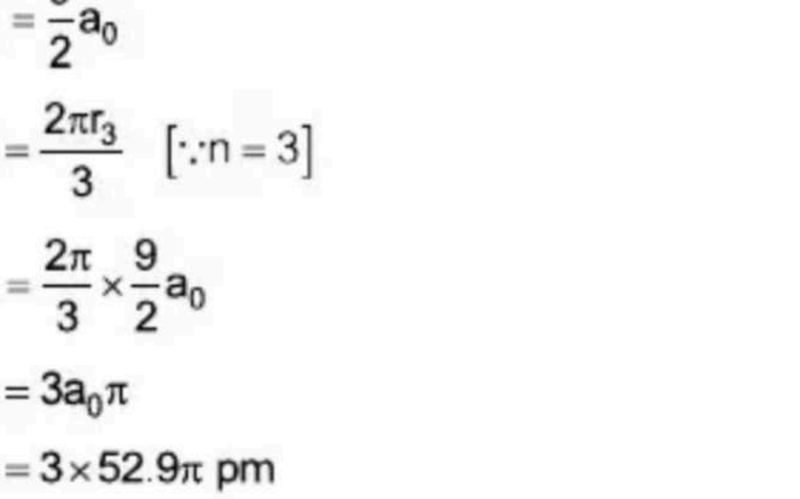
Angular momentum = $\sqrt{1(1+1)}h$

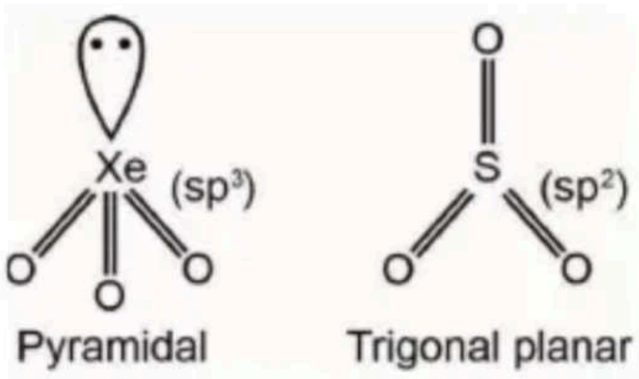
$= \sqrt{2}h$

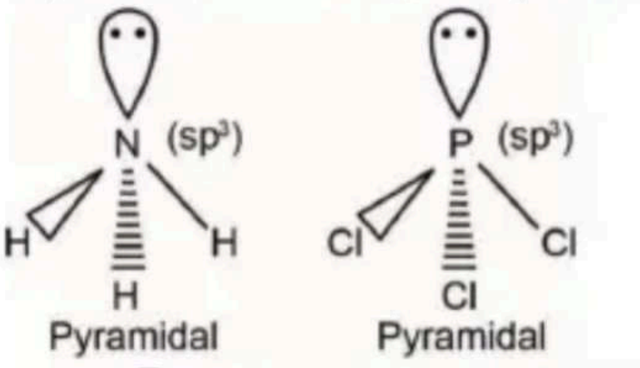
59. Answer (3)

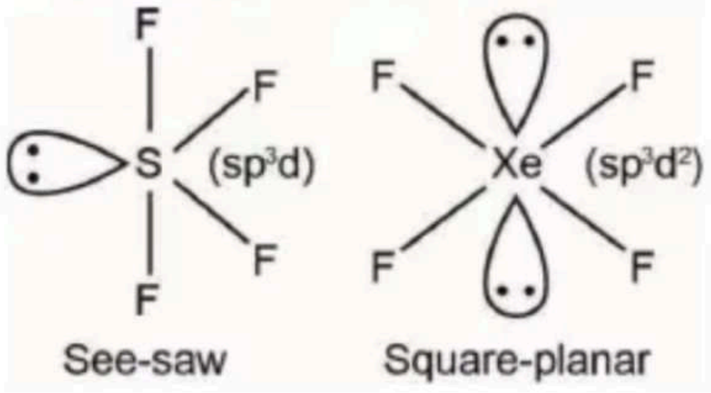
Hint: The molecules having same number of bond pairs and same number of lone pairs on central atom will be isostructural.

Sol.:

- 

- 

- 

- 

Same number of bond pairs and lone pair are present on central atom in NH_3 and PCl_3 hence they are isostructural.

60. Answer (2)

Hint: $d = \frac{PM}{RT}$

Sol.: $d = \frac{4.1 \times 32}{0.082 \times 400}$

d = 4 g/L

density of O_2 gas = 4 gL^{-1}.

61. Answer (4)

Hint: Since unit of rate constant is s^{-1} hence it is a first order reaction.

For first order reaction, $k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$

Sol.: $t = \frac{2.303}{k}\log\frac{[R]_0}{[R]}$

$t = \frac{2.303}{9.212 \times 10^{-3}}\log\frac{100}{80}$

$t = \frac{2.303}{9.212 \times 10^{-3}}[\log 10 - 3\log 2]$

$t = \frac{2.303}{9.212 \times 10^{-3}} \times 0.1$

t = 25 s

62. Answer (3)

Hint: Malachite, Calamine and Siderite are carbonate ores.

Sol:

Ores	Composition
Malachite	$CuCO_3.Cu(OH)_2$
Calamine	$ZnCO_3$
Siderite	$FeCO_3$
Haematite	Fe_2O_3

63. Answer (2)

Hint: Higher is the number of unpaired electrons, greater is the value of spin only magnetic moment.

Sol.:

Complex ion	Number of unpaired electrons (n)	$\mu=\sqrt{n(n+2)}$ BM
$[Co(H_2O)_6]^{3+}$	0	0
$[MnCl_6]^{3-}$	4	4.9
$[Fe(CN)_6]^{3-}$	1	1.7
$[Mn(CN)_6]^{3-}$	2	2.8

64. Answer (2)

Hint: HF has exceptionally high boiling point because of the presence of intermolecular hydrogen bonding.

Sol.:

Compound	Boiling point (K)
HF	293
HCl	189
HBr	206
HI	238

65. Answer (3)

Hint: 1 Faraday electricity deposits 1 g-equivalent of substance at electrode.

Sol.: Number of Faraday passed

$$= \frac{3.86\times100\times60}{96500} = 0.24\ F$$

Mass of nickel deposited = $0.24\times\frac{58.7}{2}$

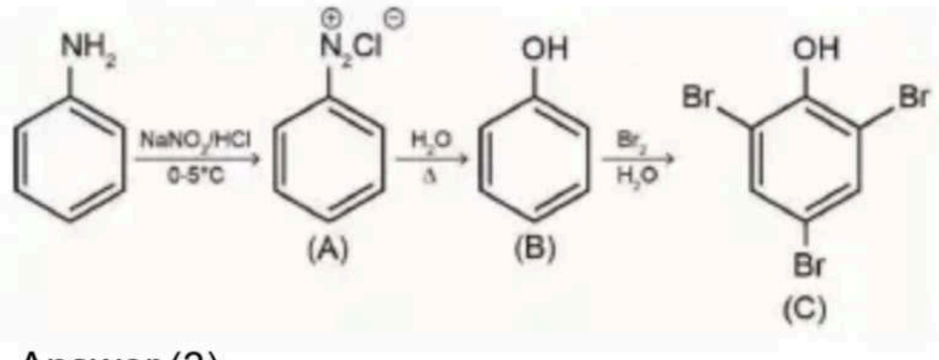
= 7.04 g

$\simeq 7$ g

66. Answer (4)

Hint: Benzene diazonium chloride is converted to phenol on reaction with water

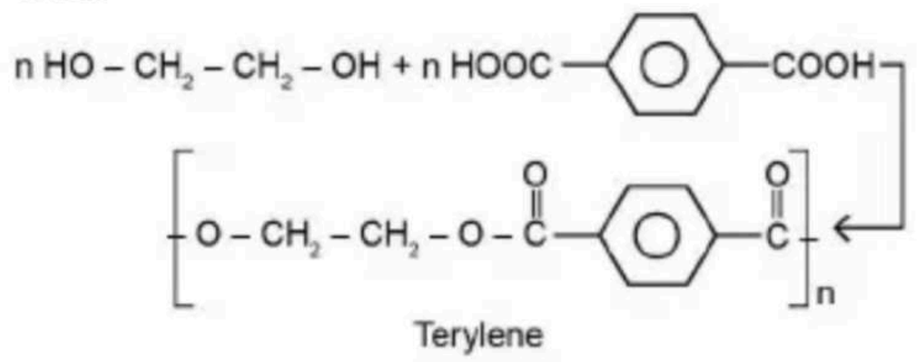
Sol.:



67. Answer (3)

Hint: Condensation polymerisation occurs between two bi-functional or trifunctional monomeric units. In this process loss of simple molecules such as water, alcohol, hydrogen chloride etc. takes place.

Sol.:



- Terylene is a condensation polymer.
- Buna-S, Teflon and Neoprene are addition polymers.

68. Answer (1)

Hint: Tranquilizers and analgesics are neurologically active drugs.

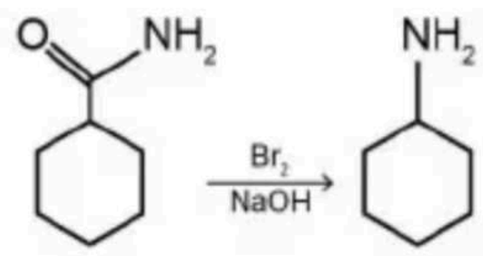
Sol.:

Drugs	Classes of drugs
Meprobamate	Neurologically active drug (Tranquilizer)
Terfenadine	Antihistamine
Tetracycline	Antibiotic
Novestrol	Antifertility drug

69. Answer (4)

Hint: The given reaction is Hoffmann bromamide degradation reaction.

Sol.:



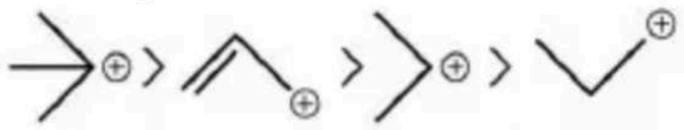
Amine formed in this reaction is one carbon less than the given amide.

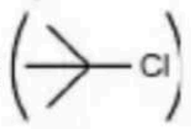
70. Answer (4)

Hint: More stable is the carbocation, faster is the S_N1 rate

Sol.:

- The intermediate species formed in the S_N1 reaction is a carbocation, More stable is the carbocation faster is S_N1 rate.
- Stability order of carbocation



- *tert* butyl chloride  will react fastest by S_N1 reaction.

71. Answer (4)

Hint: Electron withdrawing group present at para-position increases acidity of phenol derivative.

Sol.:

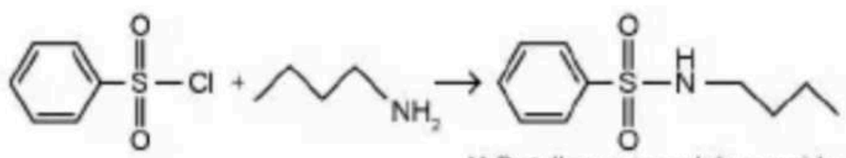
- Electron donating groups like-CH_3, -OMe decrease acidity of phenol.
- Electron withdrawing group present at para position increases acidity of phenol.
- $-NO_2$ stabilises the conjugate base by $-R$ effect hence p-nitrophenol is most acidic among the given molecules.

72. Answer (1)

Hint: Benzenesulphonyl chloride reacts with primary and secondary amines to form sulphonamides.

Sol.:

- The reaction of benzenesulphonyl chloride with primary amine yields N-alkylbenzensulphonamide. The hydrogen attached to nitrogen in sulphonamide is strongly acidic due to the presence of strong electron with drawing sulphonyl group. Hence, it is soluble in alkali



N-Butylbenzenesulphonamide
(Soluble in alkali)

- In reaction with secondary amine, N, N-dialkyl-benzenesulphonamide is formed which does not contain any hydrogen atom attached with nitrogen atom. It is not acidic and hence insoluble in alkali.
- Tertiary amine does not react with benzene sulphonyl chloride.

73. Answer (3)

Hint: Sucrose is a non-reducing sugar.

Sol.: • Sucrose on hydrolysis gives equimolar mixture of D-(+)-glucose and D-(–)–fructose

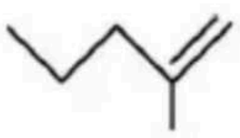
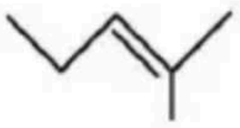
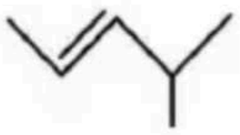
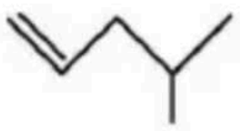
$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \rightarrow \underset{\text{D–(+)–Glucose}}{C_6H_{12}O_6} + \underset{\text{D–(–)–Fructose}}{C_6H_{12}O_6}$$

- Since the reducing groups of glucose and fructose are involved in glyosidic bond formation, hence sucrose is a non reducing sugar. It does not reduce Tollens' reagent

74. Answer (4)

Hint: More is the number of hyperconjugative structures, or α - hydrogens more is the stability of alkene and lesser is the energy released on catalytic hydrogenation.

Sol.:

Compounds	α-hydrogen atoms
	5
	8
	4
	2

75. Answer (2)

Hint: Percentage of

$$S = \frac{\text{Molar mass of S} \times \text{Mass of } BaSO_4 \times 100}{\text{Molar mass of } BaSO_4 \times \text{Mass of organic compound}}$$

Sol.: Percentage of sulphur = $\frac{32 \times 0.699 \times 100}{233 \times 0.5}$

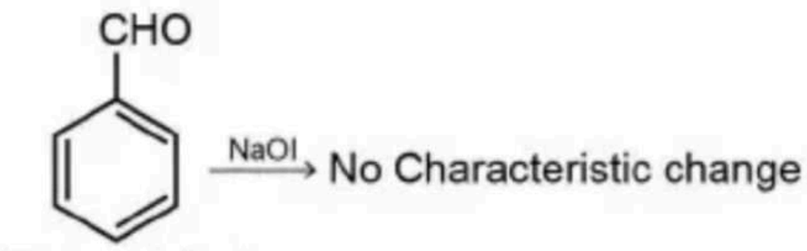
= 19.2%

76. Answer (2)

Hint: Compound containing $CH_3-\overset{\overset{O}{\|}}{C}-$ and $CH_3\underset{\underset{OH}{|}}{CH}-$ groups react with NaOI to give iodoform.

Sol.:

$$\underset{\text{Acetone}}{CH_3-\overset{\overset{O}{\|}}{C}-CH_3} \xrightarrow{NaOI} CH_3-\overset{\overset{O}{\|}}{C}-O^{\ominus} + CHI_3 \text{ (Yellow precipitate)}$$

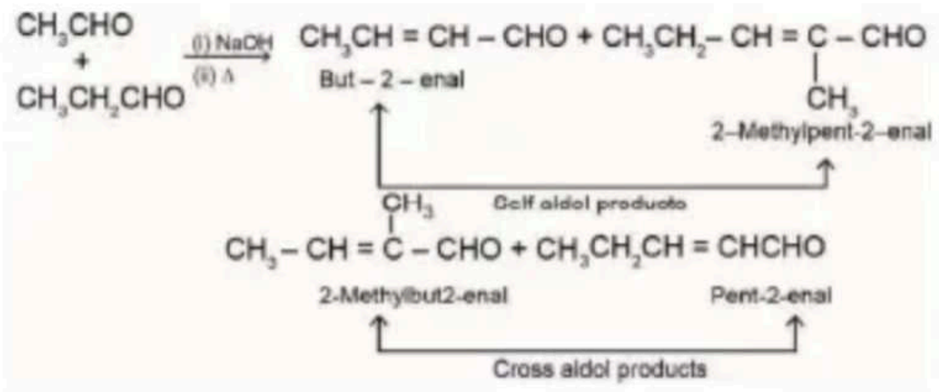


Benzaldehyde

77. Answer (2)

Hint: In a reaction of two different carbonyl compounds self aldol and cross aldol condensation reactions will take place.

Sol.:



$CH_3 - CH_2 - \underset{\underset{CH_3}{|}}{C} = CH - CHO$ will not be formed.

78. Answer (3)

Hint: The species which does not contain *d* electron(s) will not show colour in aqueous medium.

Sol.: Cu^{2+}, V^{3+} and Ni^{2+} contain *d* electrons hence these are coloured species in aqueous medium while Sc^{3+} does not contain any *d* electron. Hence Sc^{3+} is colourless in aqueous medium.

79. Answer (2)

Hint: The species which contains paired electron are diamagnetic in nature.

Sol.:

Compound /Ion	Shape & hybridisation	Magnetic behaviour
$[PtCl_4]^{2-}$	Square planar and dsp^2	Diamagnetic
$[Ni(CO)_4]$	Tetrahedral and sp^3	Diamagnetic
$[Cu(NH_3)_4]^{2+}$	Square planar and dsp^2	Paramagnetic
$[Pt(NH_3)_2Cl_2]$	Square planar and dsp^2	Diamagnetic

80. Answer (2)

Hint: $\Delta_r G^\circ = -nFE^\circ_{cell}$

Sol.: $E^\circ_{cell} = E^\circ_{Cu^{2+}/Cu} - E^\circ_{Al^{3+}/Al}$

$E^\circ_{cell} = 0.34 + 1.66$

$E^\circ_{cell} = 2V$

$\Delta_r G^\circ = -6 \times 96500 \times 2$

$= -1158$ kJ mole^{-1}

81. Answer (4)

Hint: $4Zn + 10HNO_3$ (dilute) $\rightarrow 4Zn(NO_3)_2 + 5H_2O + N_2O$

Sol.: $3Cu + 8HNO_3$ (dilute) $\rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

82. Answer (4)

Hint: • The process is adiabatic, hence q = 0

• $w = -P_{ex}\Delta V$

Sol.: $w = -5(v_f - v_i)$

$= -5(3.25 - 1.25)$

$w = -10$ L atm

From first law of thermodynamics

$\Delta U = q + w$

$\Delta U = w$ (for adiabatic process)

$\Delta U = -10$ L atm

$= -10 \times 101.3$

$\Delta U = -1013$ J

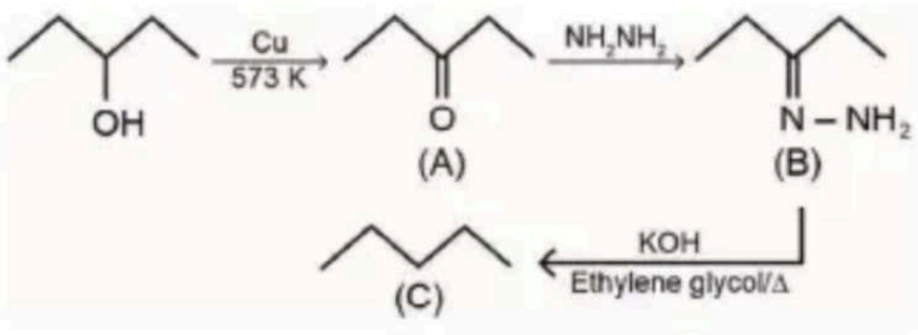
83. Answer (4)

Hint and Sol: Maximum permissible limit of lead in drinking water is 50 ppb.

84. Answer (3)

Hint: Secondary alcohol is oxidised to ketone by heating it with copper at 573 K.

Sol.:



85. Answer (1)

Hint: Amphoteric oxides behave as acid with bases and base with acids, whereas neutral oxides have no acidic or basic properties.

Sol.:

Compounds	Chemical nature
CO	Neutral
Na_2O	Basic
Cl_2O_7	Acidic
Al_2O_3	Amphoteric

SECTION-B

86. Answer (3)

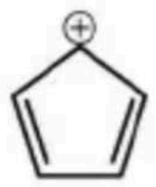
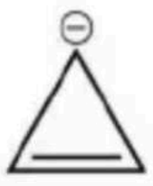
Hint: Root mean square speed (v_{rms}) = $\sqrt{\frac{3RT}{M}}$

Sol.: $\frac{v_{rms}(O_2)}{v_{rms}(He)} = \sqrt{\frac{M_{He}}{M_{O_2}}} = \sqrt{\frac{4}{32}} = 1:2\sqrt{2}$

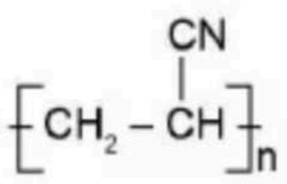
87. Answer (2)

Hint: Cyclic species containing $(4n + 2)\pi$ electrons in conjugation are aromatic in nature.

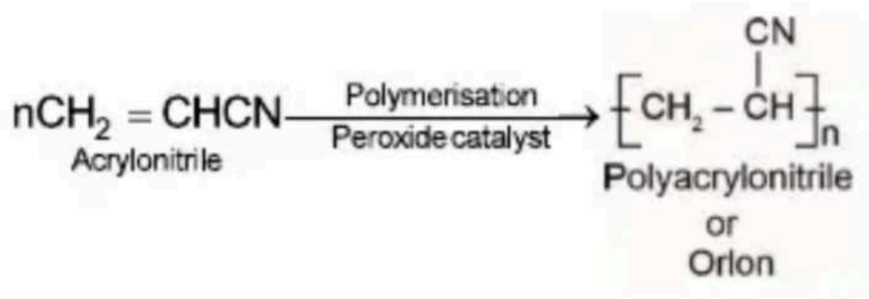
Sol.:

- Tropylium cation  contains 6π electrons in conjugation which follows Huckel's $(4n + 2)\pi$ electron rule.
-  and  are antiaromatic
-  is non-aromatic

88. Answer (3)

Hint: Orlon is 

Sol.:



89. Answer (2)

Hint: At constant pressure, on introduction of inert gas, volume increases hence equilibrium moves in a direction where number of moles of gases increases.

Sol.: At constant pressure, introduction of inert gas (helium) will lead to the increase in volume hence dissociation of ammonia increases.

90. Answer (2)

Hint: The element in its lowest or in its highest oxidation state does not undergo disproportionation reaction.

Sol.: Among the given oxoanions of chlorine, ClO_4^- does not disproportionate because in this oxoanion, chlorine is present in its highest oxidation state that is +7.

91. Answer (2)

Hint: Volume strength of H_2O_2 = 11.2 × Molarity (M)

Sol.: 22.4 = 11.2 × M

M = 2

$\left(\frac{w}{V}\right)$ % of $H_2O_2 = \frac{34\times2}{1000}\times100 = 6.8\%$

92. Answer (4)

Hint and Sol.:

Element	Flame colour
Sodium	Yellow
Caesium	Blue
Calcium	Brick red
Barium	Apple green

93. Answer (4)

Hint: Borax is a salt of weak acid and strong base.

Sol.:

- Borax is a salt of weak acid and strong base hence its aqueous solution will be alkaline in nature.

  $Na_2B_4O_7 + 7H_2O \rightarrow 2NaOH + 4H_3BO_3$

- Borax on reaction with dilute HCl gives boric acid as one of the products.

  $Na_2B_4O_7 + 2HCl + 5H_2O \rightarrow 2NaCl + 4H_3BO_3$

94. Answer (4)

Hint: Smaller is the size of cation, greater is the hydration enthalpy.

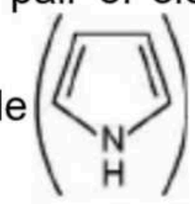
Sol.:

- In the given metal sulphates, size of Mg^{2+} is smallest. Greater hydration enthalpy of Mg^{2+} ions overcome the lattice enthalpy factor hence $MgSO_4$ is highly soluble in water.
- The solubility decreases as the size of metal cation increases.

95. Answer (3)

Hint: Lone pair present on sp^3 hybridised nitrogen atom will be most easily available for protonation.

Sol.:

- Lone pair of electrons present on nitrogen of pyrrole  are involved in resonance and the molecule becomes aromatic hence it is least basic.
- Lone pair of electrons present on the nitrogen of aniline is delocalised in the benzene ring hence its basicity decreases.
- In pyridine  the lone pair is present on sp^2 hybridised nitrogen while in benzylamine  the lone pair of electron is present on sp^3 hybridised nitrogen atom hence benzyl amine is most basic among the given options.

96. Answer (3)

Hint: Since unit of rate constant is mol $L^{-1}s^{-1}$ hence order of reaction is zero.

Sol.:

	X	→	Y
Initial	3 M		0
Final	3-kt		kt

Concentration of Y = kt

$= 2.5 \times 10^{-4} \times 100 \times 60$

$= 1.5$ M

97. Answer (4)

Hint: Sodium acetate will undergo hydrolysis in water and pH of the solution will be determined as

$$pH = 7 + \frac{1}{2}[pKa + \log c]$$

Sol.: $pH = 7 + \frac{1}{2}[4.76 + \log 0.2]$

$$pH = 7 + \frac{1}{2}[4.76 - 0.7]$$

pH = 9

98. Answer (1)

Hint: Enthalpy of neutralization of a strong base with a strong acid is -57.1 kJ eq^{-1}

Sol.:

	HCl	+ NaOH	→ NaCl + H_2O
Initial :	400×0.5	600×0.2	
	= 200 mmol	120 mmol	
Final	$(200 - 120)$		120 mmol
	= 80 mmol		

NaOH is limiting reagent

∴ Heat released = $120 \times 10^{-3} \times 57.1$

= 6.85 kJ

99. Answer (1)

Hint: Copper sol is negatively charged sol.

Sol.: A given sol is coagulated by oppositely charged ion. Copper sol is a negatively charged sol hence it is coagulated by positively charged ion. Higher is the value of charge more easily the sol will be coagulated.

100. Answer (2)

Hint: Thionyl chloride is $SOCl_2$

Sol.: $P_4 + 8SOCl_2 \longrightarrow 4PCl_3 + 4SO_2 + 2S_2Cl_2$

[BOTANY]

SECTION-A

101. Answer (1)

Hint: A pure line plant is a true-breeding plant which shows stable trait expression and inheritance for several generation due to self-pollination

Sol.: Mendel choose 7 pairs of contrasting traits in pea plants.

102. Answer (2)

Hint: Homologous chromosomes pair before the crossing over process which occurs in pachytene

Sol.: Leptotene – Condensation of chromosomes.

Zygotene – Pairing of homologous chromosomes.

Pachytene – Crossing over.

Diplotene – Separation of homologous chromosomes.

103. Answer (2)

Hint: Cells of this region differentiate to form certain structures.

Sol.: The epidermal cells from the zone of maturation form fine and delicate, thread-like structures called root hairs.

104. Answer (2)

Hint: Heterostyly refers to difference in the length of stamen filament and style of stigma in a bisexual flower.

Sol.: It is a morphological method in some plants like *Primrose* to prevent self pollination.

105. Answer (1)

Hint: In one turn of Krebs cycle NAD^+ and FAD^+ are reduced.

Sol.: In one turn of Krebs cycle, 3 molecules of NAD^+ are reduced to NADH + H^+ and one molecule of $FADH_2$ is synthesized.

106. Answer (1)

Hint: RNA polymerase synthesizes mRNA.

Sol.: DNA polymerase can polymerise 2,000 bp per second only in one direction i.e., $5' \rightarrow 3'$ direction.

107. Answer (2)

Hint: In parasitism and predation, one species is benefitted and another one is harmed.

Sol.: Plants and fungi in mycorrhiza share mutualistic relationship.

Eagle is a predator of rat.

Penicillium produces antibiotic penicillin, which kills many bacteria.

Lice are ectoparasites on humans.

108. Answer (3)

Hint: Along with water, dissolved minerals are absorbed by roots and translocated by xylem.

Sol.: Xylem translocates water, absorbed minerals, fixed nitrogen (as nitrates) and hormones. Photosynthates are transported by phloem.

109. Answer (3)

Hint: Phylogenetic system of classification was proposed by Engler and Prantl, Hutchinson and Takhtajan.

Sol.: Mosses (bryophytes) show haplo-diplontic life cycle.

Eudorina is an alga in which two types of gametes are produced.

In brown algae, the food is stored in the form of laminarin or mannitol.

110. Answer (3)

Hint: The pigment that traps sunlight for light reaction of photosynthesis has a tadpole like structure.

Sol.: Photorespiration is a wasteful process as it involves loss of fixed carbon.

Ruben and Kamen proved that the O_2 liberated during photosynthesis comes from water.

Kranz anatomy is observed in C_4 plants.

111. Answer (3)

Hint: This chromosome appears J-shaped during anaphase.

Sol.: One arm of the acrocentric chromosome is very small and one very long, gives it a J-shaped appearance during anaphase.

112. Answer (2)

Hint: Hypodermis is absent in roots.

Sol.: Conjoint and open vascular bundles, presence of medullary rays and pith, and collenchymatous hypodermis are the characteristics of dicot stem.

113. Answer (1)

Hint: NAD^+ is a co-enzyme which accepts electrons during different steps of respiration.

Sol.: NAD^+ acts as an electron carrier by transporting electrons to the ETS.

114. Answer (1)

Hint: Vegetative propagation in ginger occurs through underground stem.

Sol.: Eye of tuber – Potato

Leaf buds – *Bryophyllum*

Offset – Water hyacinth

Rhizome – Ginger

115. Answer (4)

Hint: In the most tedious step of plant breeding, the pollen grain from male parent has to be collected and placed on the stigma of female parent, manually.

Sol.: Cross-hybridisation among the selected parents is performed which is very tedious and time-consuming process.

116. Answer (2)

Hint: The pioneer species in hydrarch succession are microscopic autotrophic organisms.

Sol.: Phytoplanktons are the pioneer species in primary succession in water, followed by rooted-submerged, submerged free-floating plants, reed-swamp stage, marsh-meadow stage, scrub stage and finally forest.

117. Answer (3)

Hint: In case of female heterogamety, ZW-ZZ type of sex determination is seen.

Sol.: Grasshopper and *Drosophila* show male heterogamety.

In birds, females produce heterogametes A + Z and A + W.

118. Answer (2)

Hint: Ribosomes bound to ER perform protein synthesis.

Sol.: RER → Protein synthesis

SER → Lipids synthesis

Golgi complex → Glycosylation

Centrioles → Spindle formation in cell division

119. Answer (2)

Hint: Mortality reduces population density of an area.

Sol.: Mortality is the rate of deaths in a population.

120. Answer (1)

Hint: It nourishes the developing pollen grains.

Sol.: Epidermis, endothecium and middle layers are protective in function while tapetum provide nourishment.

121. Answer (2)

Hint: It is absorbed in oxidation state of +2

Sol.: Copper is absorbed as cupric (Cu^{2+}) ions by plants.

122. Answer (2)

Hint: Bad ozone is present in Troposphere.

Sol.: Good ozone is present in Stratosphere.

123. Answer (3)

Hint: This hormone is responsible for apical dominance in plants.

Sol.: Auxin was isolated from human urine.

124. Answer (3)

Hint: Tropical regions have higher species diversity than temperate regions.

Sol.: Tropical Amazonian rain forest has the greatest biodiversity on earth with >40,000 species of plants, 3000 species of fishes, 1300 species of birds, 427 species of mammals, etc.

125. Answer (2)

Hint: Viroids are made up of RNA only.

Sol.: Prions cause mad cow disease in cattle and Cr-Jacob disease in humans. Their size is similar to that of viruses.

126. Answer (1)

Hint: *Chlamydomonas* and *Volvox* are unicellular and colonial algae, respectively.

Sol.: *Ulothrix* and *Spirogyra* are filamentous algae.

127. Answer (2)

Hint: Dumb-bell shaped guard cells are present in grasses.

Sol.: Guard cells are usually kidney or bean shaped in dicot leaves.

128. Answer (4)

Hint: Acetic acid is obtained from of a bacterium

Sol.: *Penicillium* → Penicillin

Aspergillus → Citric acid

Lactobacillus → Lactic acid

Acetobacter → Acetic acid

129. Answer (1)

Hint: Probes used in DNA fingerprinting are made up of repetitive DNA.

Sol.: Variable Number of Tandem Repeats (VNTRs) are used as probe for DNA fingerprinting.

130. Answer (4)

Hint: In grassland ecosystems, the ecological pyramids are upright

Sol.: Pyramid of biomass in sea is inverted because the number of zooplanktons (consumers) is usually much higher than that of producers (phytoplanktons).

131. Answer (3)

Hint: Cork is formed by the differentiation of cells produced by cork cambium on the outer side and is protective in function.

Sol.: Due to origin of cork cambium outside the stele, it is also called the extra-stelar cambium.

132. Answer (3)

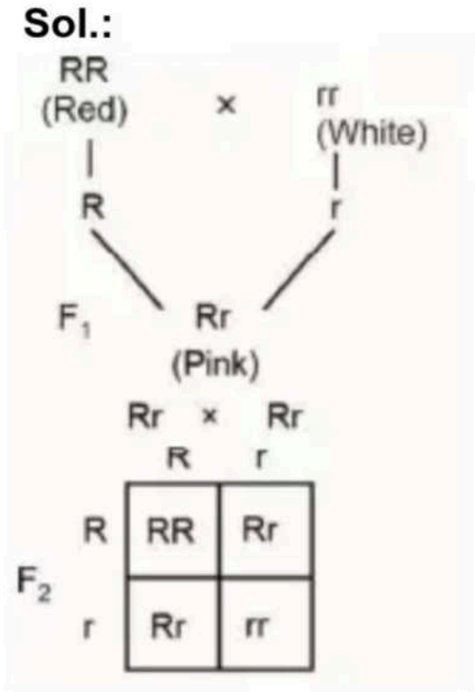
Hint: In the S-phase, a non-membrane bound cell organelle is duplicated.

Sol.: In S-phase, DNA is duplicated in the nucleus and centrioles in the cytoplasm.

133. Answer (2)

Hint: In *Antirrhinum,* incomplete dominance is seen.

Sol.:



Red : Pink : White = 1 : 2 : 1

134. Answer (3)

Hint: Some algae and fungi can be in a symbiotic association.

Sol.: A = Lichens, composed of phycobiont and mycobiont, and it cannot grow in polluted environment.

B = Viruses, inside host they act as living and outside them they act as non-living.

135. Answer (3)

Hint: Bacterial blight is a disease of Cowpea.

Sol.: Pusa Komal variety of cow pea is resistant to bacterial blight.

SECTION-B

136. Answer (3)

Hint: Each statement of the contrasting pair in a key is called a lead.

Sol.: The keys are based on the contrasting character generally in a pair called couplet.

137. Answer (2)

Hint: Ribulose-1, 5-biphosphate is the primary acceptor of CO_2 in Calvin cycle.

Sol.: Hatch and Slack pathway is observed in C_4 plants. Cyclic photophosphorylation involves only PS I.

138. Answer (3)

Hint: Zoospores are meant for asexual reproduction.

Sol.: Zoospores are produced by mitosis in *Chlamydomonas* for asexual reproduction. Megaspore cells are produced by meiotic division of megaspore mother cell.

139. Answer (4)

Hint: Colourful flowers and nectar are present in insect-pollinated plants.

Sol.: The pollen grains of the water-pollinated flowers are protected from wetting by mucilaginous covering.

140. Answer (4)

Hint: World Summit of 2002 at Johannesburg involved 190 countries, who took a pledge for reducing loss of biodiversity.

Sol.: Reducing the rate of biodiversity loss – World Summit, 2002

Reducing emission of Greenhouse gases – Kyoto protocol

Protecting the stratosphere – Montreal protocol

Conservation of Biodiversity and Sustainable utilization of its benefits – Earth Summit, Rio De Janeiro, 1992

141. Answer (2)

Hint: Histone protein synthesis takes place in S-phase of interphase. Longest stage of Meiosis I is Prophase I.

Sol.: Duplication of DNA → S-phase

Longest phase of cell cycle → Interphase

Longest stage of meiosis I → Prophase I

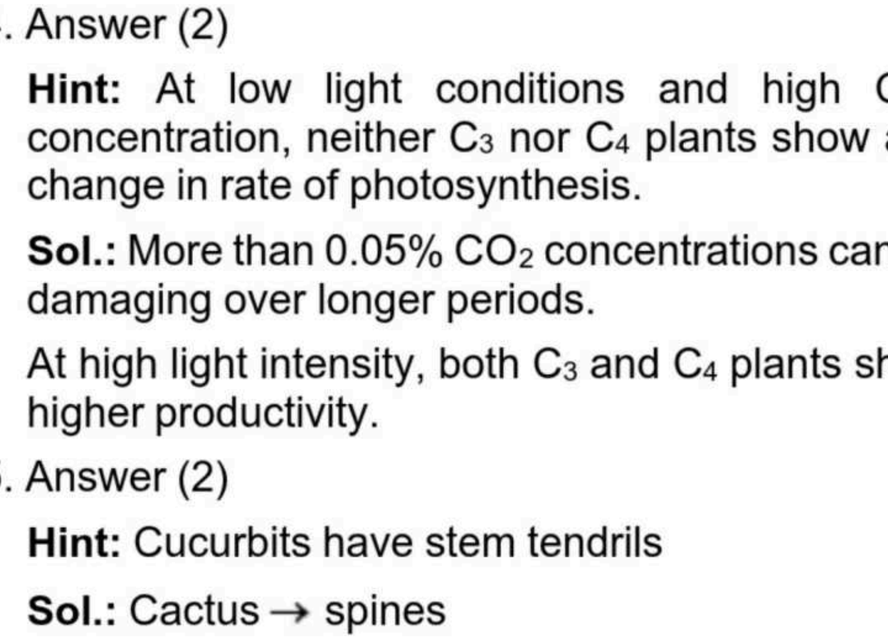
Terminalisation of chiasmata → Diakinesis

142. Answer (3)

Hint: Colour blindness is a sex linked recessive disorder.

Sol.: Colourblind female: X^cX^c

Normal male: XY



All the females are carrier of this disease *i.e.*, no female progeny is colourblind.

143. Answer (3)

Hint: The events of ETS occur only in the mitochondria.

Sol.: During the transportation of electrons in the ETS, the protons move to intermembrane space from mitochondrial matrix.

144. Answer (2)

Hint: At low light conditions and high CO_2 concentration, neither C_3 nor C_4 plants show any change in rate of photosynthesis.

Sol.: More than 0.05% CO_2 concentrations can be damaging over longer periods.

At high light intensity, both C_3 and C_4 plants show higher productivity.

145. Answer (2)

Hint: Cucurbits have stem tendrils

Sol.: Cactus → spines

Pisum sativum → leaf tendril

Members of Cucurbitaceae → stem tendril

Bryophyllum → leaf buds

146. Answer (3)

Hint: Females suffering from this condition show monosomy of sex chromosome.

Sol.: Down's syndrome: Trisomy at chromosome 21

Klinefelter's syndrome : 44 + XXY

Turner's syndrome : 44 + XO

Edwards syndrome : Trisomy at chromosome 18.

147. Answer (3)

Hint: Statins are produced by a fungus.

Sol.: *Trichoderma polysporum* – Cyclosporin A

Streptococcus – Streptokinase

Monascus purpureus – Statins

Clostridium – Butyric acid

148. Answer (2)

Hint: RNA polymerase III transcribes tRNA, 5S rRNA and snRNAs.

Sol.: RNA polymerase II transcribes hnRNA only. RNA polymerase I transcribes 28S, 18S and 5.8S rRNAs.

149. Answer (2)

Hint: Most of the organelles duplicate in G_1 phase and centrioles duplicate in S-phase.

Sol.: Tubulin protein is synthesized in G_2 phase. Only the organelles like mitochondria, chloroplasts and Golgi complex duplicate in this phase.

150. Answer (3)

Hint: Dormancy is broken by ethylene and auxins lead to apical dominance.

Sol.: Gibberellin – Helps in seed germination

Auxins – Present in root and shoot apex

Abscissic acid – Shows inhibitory effect

Cytokinin – Overcomes apical dominance

[ZOOLOGY]

SECTION-A

151. Answer (2)

Hint: Connects the cytoplasm of adjoining cells

Sol.: Three types of cell junctions are found in the epithelium and other tissues.

1. Tight junctions – Help to stop substances from leaking across a tissue
2. Adhering junctions – Perform cementing to keep neighbouring cells together
3. Gap junctions – Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells for rapid transfer of ions, small molecules, *etc*.

152. Answer (4)

Hint: Inner cell mass differentiates into embryo.

Sol.: In later phase of pregnancy, a hormone called relaxin is also secreted by the ovary. hCG, hPL and relaxin are produced in women only during pregnancy. Inner cell mass contains certain cells called stem cells which have the potency to give rise to all the tissues and organs.

153. Answer (2)

Hint: Diabetes insipidus

Sol.: ADH facilitates water reabsorption from latter parts of renal tubule, thereby preventing diuresis.

Presence of proteins in urine is termed proteinuria.

154. Answer (2)

Hint: It has 16 carbons including carboxyl carbon

Sol.: Palmitic acid has 16 carbons including the carboxyl carbon.

Glutamic acid and aspartic acid are acidic amino acids.

155. Answer (2)

Hint: Female *Anopheles* mosquito is a vector of this disease

Sol.: Symptoms of amoebiasis include constipation, abdominal pain and cramps, stool with excess mucous and blood clots.

Filariasis is characterised by the inflammation of lymphatic vessels of the lower limbs.

Appearance of dry, scaly lesions on various parts of the body such as skin, nails and scalp are the main symptoms of ringworms.

156. Answer (2)

Hint: Production of 6 – 8 eggs

Sol.: MOET (Multiple Ovulation Embryo Transfer Technology) is a programme for herd improvement.

In AI, the semen is collected from the selected male and injected into the reproductive tract of the selected female by the breeders.

In cross-breeding, superior males of one breed are mated with superior females of another breed. Mating between two different related species is called interspecific hybridisation.

157. Answer (2)

Hint: Related to structure involved in haemopoiesis

Sol.: Gene therapy is an attempt as a corrective therapy of a person who was born with a hereditary disease.

cry gene is isolated from *Bacillus thuringiensis*.

ADA enzyme is crucial for the immune system to function. In some children, ADA deficiency can be cured by bone marrow transplantation.

α–interferon is a biological response modifier which activates the immune system and helps in destroying the tumor.

158. Answer (2)

Hint: One of the major cause is cigarette smoking

Sol.: Asthma is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

Diarrhoea is abnormal frequency of bowel movement and increased liquidity of the faecal discharge.

Muscular dystrophy is the progressive degeneration of skeletal muscle mostly due to genetic disorder.

159. Answer (2)

Hint: Bones of hind limb

Sol.: The vertebral column is differentiated into cervical (7), thoracic (12), lumbar (5), sacral (1-fused) and coccygeal (1-fused) regions starting from the skull.

Atlas is the first vertebra. Atlas articulates with the occipital condyles.

Ethmoid and sphenoid are cranial bones.

Sternum and ribs are axial bones which participates in formation of thoracic cage.

160. Answer (3)

Hint: Primary spermatocyte undergoes meiosis-I

Sol.:

Primary spermatocyte Primary oocyte Spermatogonia Oogonia	Diploid cells
Ootid Spermatozoa Spermatids	Haploid cells

161. Answer (3)

Hint: Exclude the anti-diuretic hormone.

Sol.: Foetal ejection reflex triggers the release of oxytocin from maternal pituitary. ACTH is a tropic hormone secreted from pars distalis.

ADH or vasopressin stimulates reabsorption of water from distal part of renal tubules.

162. Answer (3)

Hint: Related to human involvement

Sol.: Adaptive radiation leads to homology. Evolution for Darwin was gradual while de Vries believed mutation caused speciation and hence called it saltation. The process of evolution of antibiotic resistant bacteria, pesticide resistant pests, *etc*. are due to anthropogenic action.

163. Answer (3)

Hint: Low pH favours dissociation of oxyhaemoglobin

Sol.: O_2 can bind with haemoglobin in a reversible manner to form oxyhaemoglobin. Low pO_2, high pCO_2, high H^+ concentration and high temperature are favourable conditions for dissociation of oxygen from oxyhaemoglobin. This clearly indicates that O_2 gets bound to haemoglobin in the lung surface and gets dissociated at the tissues. Oxyhaemoglobin dissociation curve shifts to right on increasing temperature, pCO_2, H^+ and BPG.

164. Answer (2)

Hint: Progestasert belongs to the same category

Sol.: IUDs increase phagocytosis of sperms within the uterus and the copper ions released supress sperm motility and the fertilising capacity of sperms.

Lippes loop and Multiload-375 are non-medicated and Cu releasing IUDs, respectively.

165. Answer (3)

Hint: A reduction in number of thrombocytes leads to clotting disorders

Sol.: Platelets, also called thrombocytes, are cell fragments produced from megakaryocytes. Blood normally contains 1,500,00 – 3,500,00 platelets mm^{-3}. Platelet factors and tissue factors interact in presence of Ca^{+2} to form an enzyme complex thrombokinase which converts prothrombin into active thrombin.

166. Answer (3)

Hint: Metameric segmentations are seen in *Pheretima*

Sol.: The canal system in sponges is helpful in food gathering, respiratory exchange and removal of wastes.

In metamerism, the body is externally and internally divided into segments with a serial repetition of at least some organs.

In ctenophores, body bears 8 external rows of ciliated comb plates.

167. Answer (2)

Hint: According to Chargaff rule,

A = T, C = G

Sol.: Out of 500 nucleotide base pairs

i.e., 500 × 2 = 1000 nucleotides

C = 40%

So, G = 40%

A + T % = 100 – [G + C]%

= 100 – 80 = 20%

A + T = 20%

∵ A = T so, A = 10% of total nucleotides

$$A = \frac{1000 \times 10}{100} = 100$$

168. Answer (4)

Hint: Cystic duct along with hepatic ducts form common hepatic duct

Sol.: Brunner's glands are submucosal glands.

The bile duct and the pancreatic duct open together into the duodenum as the common hepato-pancreatic duct which is guarded by sphincter of Oddi.

169. Answer (1)

Hint: In barrier methods, ovum and sperms are prevented from physical meeting

Sol.: Diaphragms, cervical caps, and vaults prevent conception by blocking the entry of sperms through the cervix. Surgical methods (sterilization) are generally advised for the male/female partner as a terminal method to prevent any more pregnancies.

170. Answer (4)

Hint: Roundworms

Sol.: Aschelminthes exhibit bilateral symmetry. Adult echinoderms, cnidarians and ctenophores exhibit radial symmetry.

171. Answer (1)

Hint: Present in hemichordates

Sol.: Proboscis glands are reported in hemichordates. Metanephric kidney, 4-chambered heart and presence of feathers are features of birds.

172. Answer (3)

Hint: Class VI of enzymes

Sol.: Restriction endonuclease cut DNA at specific site (their sugar phosphate backbone).

The resulting DNA fragments will have complementary overhanging stretches (sticky ends). Flush end is also called blunt end.

173. Answer (4)

Hint: Restriction endonuclease was isolated from it in 1963

Sol.: Plasmid of *E.coli* is used in production of humulin.

Thermus aquaticus is a thermophilic bacteria and *Taq* polymerase is used in PCR.

Meloidogyne incognita (nematode) infects the roots of tobacco plants and causes a great reduction in yield.

174. Answer (4)

Hint: Used for treatment of cancer

Sol.:

PCR	–	Detect HIV in suspected AIDS patients
ELISA	–	Detecting presence of antigens or antibodies synthesized against pathogen
Radiotherapy	–	Used to treat cancer
Radiography	–	Diagnosis of cancer

175. Answer (4)

Hint: Diabetes mellitus

Sol.: PCR, ELISA and rDNA technology serves the purpose of early diagnosis of diseases. Hepatitis-B and HIV are viral STDs.

Mature insulin has two polypeptide chains.

176. Answer (2)

Hint: Identify a domestic animal

Sol.: *Corvus* is a bird and *Chelone* is a reptile.

Ascidia is a protochordate and *Canis* is a mammal.

177. Answer (2)

Hint: Identify an amino acid

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc*.
Alkaloids	Morphine, Codeine, *etc*.
Terpenoides	Monoterpenes, Diterpenes *etc*.

Essential oils	Lemon grass oil, *etc*.
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc*.
Polymeric substances	Rubber, gums, cellulose

178. Answer (1)

Hint: Prevents polyspermy

Sol.: The secretions of the acrosome help the sperm enter into the cytoplasm of the ovum through the zona pellucida and the plasma membrane. Sperm attaches with zona pellucida layer of ovum and induces stabilization of membrane to prevent further entry of sperms.

179. Answer (2)

Hint: Blood group devoid of antigens A and B

Sol.:

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

180. Answer (2)

Hint: May lead to paralysis of skeletal muscles

Sol.: Osteoporosis is characterized by decreased bone mass and increased chances of fractures.

Myasthenia gravis – Fatigue, weakening and paralysis of skeletal muscles.

Gout occurs due to accumulation of uric acid crystals in joints.

181. Answer (2)

Hint: Movement of ovum in fallopian tube is not a voluntary action.

Sol.: Smooth muscles (non-striated/visceral muscles) are not under the voluntary control of the nervous system and are therefore known as involuntary muscles.

182. Answer (2)

Hint: Bone resorption

Sol.: Along with TCT, it plays a significant role in calcium balance in the body.

PTH acts on bones and stimulates the process of bone resorption (dissolution/demineralisation).

183. Answer (1)

Hint: Commonly known as sea fan

Sol.: *Limulus* (arthropod), *Pinctada* (mollusc) and *Nereis* (annelid) are triploblastic coelomates. *Gorgonia* is a diploblastic cnidarian.

184. Answer (3)

Hint: pCO_2 in deoxygenated blood is 45 mmHg.

Sol.:

Respiratory Gas	Atmospheric air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

185. Answer (2)

Hint: Secretion of brush border cells

Sol.: The food mixes thoroughly with the acidic gastric juice of the stomach by the churning movements of its muscular wall and is called chyme.

Pancreatic juice contains nucleases. Gastric juice contains pepsinogen and HCl.

SECTION-B

186. Answer (2)

Hint: Includes the segment containing a pair of spermatheca

Sol.: A pair of ovaries lie laterally in 2nd to 6th abdominal segments.

Anal cerci are paired, jointed out growths, which arise from the 10th tergum.

Mushroom or utricular glands are present in 6th – 7th abdominal segments of a male cockroach.

187. Answer (4)

Hint: ADH is released during dehydration condition

Sol.: ANF can cause vasodilation.

$$\text{Angiotensin I} \xrightarrow{\text{Renin}} \text{Angiotensin II}$$

Erythropoietin is produced by juxtaglomerular cells of the kidneys.

188. Answer (4)

Hint: DNA contains deoxyribose sugar.

Sol.: The sugar found in polynucleotides is either ribose or 2' deoxyribose. Proteins are polymers of amino acids. Starch forms helical structure which can hold I_2 in its helices.

189. Answer (2)

Hint: Tetracycline resistance gene has two restriction sites.

Sol.: amp^R gene in pBR322 can be cleaved by *Pvu* I and *Pst* I.

The rop codes for the proteins involved in the replication of the plasmid.

When a rDNA is injected within the coding sequence of β-galactosidase, it results into insertional inactivation.

190. Answer (1)

Hint: Lined by sarcolemma

Sol.: Each organised skeletal muscle in our body is made of a number of muscle bundles or fascicles held together by a common collagenous connective tissue layer called fascia. Each muscle bundle contains a number of muscle fibres. Each muscle fibre is lined by the plasma membrane called sarcolemma enclosing the sarcoplasm.

191. Answer (3)

Hint: Class of transferases

Sol.: Catalyses the transfer of a group, G (other than hydrogen) between a pair of substrate S and S′. Class I includes oxidoreductases, class III includes hydrolases and class IV includes lyases.

192. Answer (3)

Hint: Founder effect

Sol.: Founder effect is the example of genetic drift. Sometimes, the change in allelic frequency is so different in the new sample of population that they become a different species.

193. Answer (3)

Hint: An organisation set up by Indian government

Sol.: GEAC – Genetic Engineering Approval Committee.

European Federation of Biotechnology gave the definition of biotechnology.

194. Answer (2)

Hint: *Thermus aquaticus*

Sol.: *Taq* polymerase adds nucleotides in 5′ → 3′ direction.

PCR is a reaction that is utilized to amplify a gene or fragment of DNA of interest.

Each cycle of PCR has 3 steps

1. Denaturation : Starting solution is heated to 94°C
2. Annealing : Reaction mixture is cooled in between 50-60°C
3. Extension : Mixture is heated to 72°C

195. Answer (1)

Hint: One cannot see pure DNA fragments in the visible light

Sol.: Since DNA fragments are negatively charged molecules, they can be separated by forcing them to move towards the anode under an electric field through a medium/matrix.

196. Answer (1)

Hint: One cardiac cycle represents one heart beat.

Sol.: Normal haemoglobin in blood = 12 – 16 g/dl.

Normal blood pressure $= \frac{120}{80}$ mmHg

Duration of a cardiac cycle = 0.8 sec.

$$\text{Lub} \underset{0.5\,\text{sec}}{\overset{0.3\,\text{sec}}{\rightleftharpoons}} \text{Dub}$$

197. Answer (3)

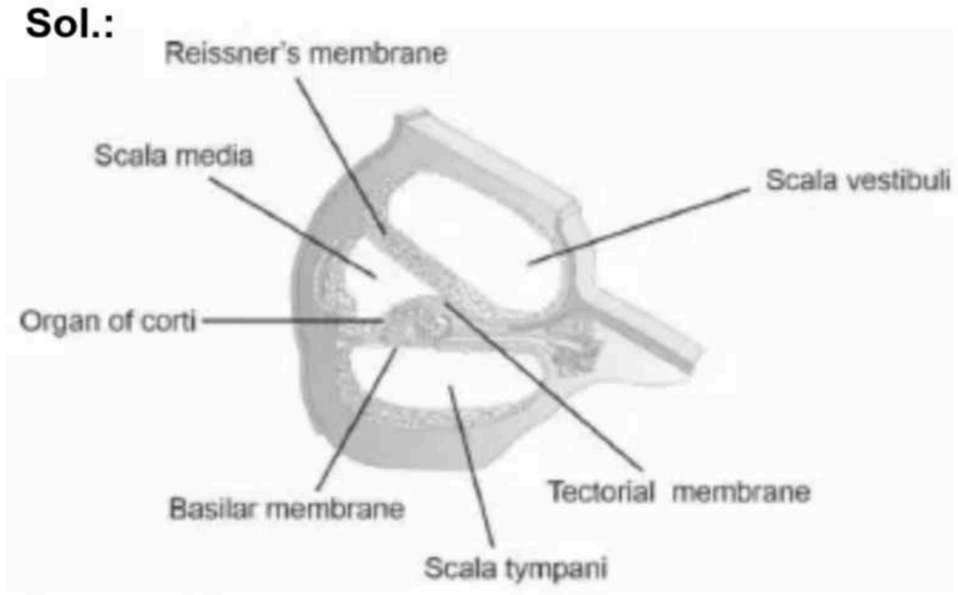
Hint: It is not a true segment

Sol.: The first body segment of earthworm called peristomium, bears mouth and a prostomium, that covers the mouth partially. Prostomium helps in formation of burrow.

198. Answer (2)

Hint: The coiled portion of the labyrinth is called cochlea

Sol.:



199. Answer (3)

Hint: Hardy-Weinberg principle – The gene pool (total genes and their alleles in a population) remains constant.

Sol.: The frequency of homozygous recessive babies $(q^2) = \frac{90}{1000} = 0.09$

$q = \sqrt{0.09} = 0.3$

$p = 1 - q = 0.7$

The frequency of heterozygous individuals = 2pq = 2 × 0.7 × 0.3 = 0.42

200. Answer (2)

Hint: Hypothalamus is associated with thermoregulation.

Sol.: The hypothalamus contains a number of centres, which control body temperature, urge for eating and drinking.

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